



**COMSATS University Islamabad,  
Sahiwal Campus**

**Customized Academic Portfolio**

**A Project Presented To**

**COMSATS University Islamabad,  
Sahiwal Campus**

**In partial fulfillment**

**of the requirement for the degree of**

**Bachelor of Science in Computer Science (2022-2026)**

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# Certificate Of Approval

It is to certify that the final year project of BS(CS) “**Customized Academic Portfolio**” was developed by **Mahum Naveed (CIIT/SP22-BCS-161/SWL)**, **Ahmad Nawaz (CIIT/SP22-BCS-183/SWL)**, **Rida Nayab (CIIT/SP22-BCS-192/SWL)** under the supervision of “**Mr. Ahmad Shaf**” and co supervisor “**Mr. Muhammad Aamir**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

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# Executive Summary

Customized Academic Portfolio (CAP) is an innovative web-based platform designed to support the academic and professional growth of fresh graduates and final-semester students who are preparing for their first internships and job opportunities. The platform offers a centralized space where students can build personalized academic portfolios, practice interviews through AI-driven mock sessions, and connect directly with recruiters who are actively seeking fresh talent.

CAP helps individuals showcase their skills, academic projects, achievements, and job readiness in a structured, professional, and visually appealing format. Unlike traditional job platforms, CAP focuses specifically on early-stage candidates and ensures they receive guidance, preparation, and visibility during their transition from academia to industry.

To support students who feel unprepared or confused during job hunting, CAP integrates an AI Mock Interview Module. This feature simulates real interview scenarios, evaluates communication and technical understanding, identifies weak areas, and provides personalized feedback to help students improve their confidence and performance.

The platform includes a Portfolio Builder, allowing users to create a complete digital academic identity with CVs, cover letters, project highlights, and interview insights. This ensures that even students without prior experience can present themselves professionally.

For recruiters, CAP offers a straightforward and efficient interface where they can browse candidate portfolios, apply filters based on skills or education, post job openings, and directly interact with potential hires. CAP does not use automated fit-scoring or ranking systems ensuring transparent, human-centered hiring.

Built using modern technologies such as Django and PostgreSQL, the system ensures secure authentication, smooth data processing, and a user-friendly experience accessible to users from all technical backgrounds. Although the platform includes AI for interview preparation, it intentionally avoids automated candidate scoring to maintain fairness and simplicity.

The project follows an Agile development model, allowing continuous improvements through real-user feedback from students, recruiters, and academic mentors. Comprehensive testing including unit, integration, system, and user acceptance testing ensures reliability, accuracy, and seamless functionality across all features.

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# Abbreviations

|                              |                                                         |
|------------------------------|---------------------------------------------------------|
| <b>UT</b>                    | Unit Testing                                            |
| <b>FT</b>                    | Functional Testing                                      |
| <b>IT</b>                    | Integration Testing                                     |
| <b>CAP</b>                   | Customized Academic Portfolio                           |
| <b>CUI</b>                   | Comsats University Islamabad                            |
| <b>UC</b>                    | Use Case                                                |
| <b>2FA</b>                   | Two-Factor Authentication                               |
| <b>AI</b>                    | Artificial Intelligence                                 |
| <b>API</b>                   | Application Programming Interface                       |
| <b>BERT</b>                  | Bidirectional Encoder Representations from Transformers |
| <b>BSc</b>                   | Bachelor of Science                                     |
| <b>CAP</b>                   | Customized Academic Portfolio                           |
| <b>CI/CD</b>                 | Continuous Integration/Continuous Deployment            |
| <b>COMSATS</b>               | COMSATS University Islamabad                            |
| <b>CSS3</b>                  | Cascading Style Sheets 3                                |
| <b>CV</b>                    | Curriculum Vitae                                        |
| <b>Django REST Framework</b> | Django REST Framework (backend API framework)           |
| <b>DRF</b>                   | Django REST Framework                                   |
| <b>FT</b>                    | Functional Testing                                      |
| <b>GDPR</b>                  | General Data Protection Regulation                      |
| <b>HTML5</b>                 | HyperText Markup Language 5                             |
| <b>IT</b>                    | Integration Testing                                     |
| <b>JavaScript</b>            | JavaScript (programming language)                       |
| <b>jQuery</b>                | jQuery (JavaScript library)                             |
| <b>KPIs</b>                  | Key Performance Indicators                              |
| <b>LightGBM</b>              | Light Gradient Boosting Machine                         |
| <b>LMS</b>                   | Learning Management System                              |
| <b>MVC</b>                   | Model-View-Controller                                   |
| <b>MVT</b>                   | Model-View-Template                                     |
| <b>NLP</b>                   | Natural Language Processing                             |

|                   |                                         |
|-------------------|-----------------------------------------|
| <b>NLTK</b>       | Natural Language Toolkit                |
| <b>NumPy</b>      | Numerical Python                        |
| <b>ORCID</b>      | Open Researcher and Contributor ID      |
| <b>Pandas</b>     | Pandas (data manipulation library)      |
| <b>PDF</b>        | Portable Document Format                |
| <b>PostgreSQL</b> | PostgreSQL (relational database)        |
| <b>RBAC</b>       | Role-Based Access Control               |
| <b>REST</b>       | Representative State Transfer           |
| <b>spaCy</b>      | spaCy (NLP library)                     |
| <b>SQLite</b>     | SQLite (lightweight database)           |
| <b>TensorFlow</b> | TensorFlow (machine learning framework) |
| <b>UI</b>         | User Interface                          |
| <b>UI/UX</b>      | User Interface/User Experience          |
| <b>UC</b>         | Use Case                                |
| <b>UT</b>         | Unit Testing                            |

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# **Chapter 1**

## **Introduction**

# 1. Introduction

The Customized Academic Portfolio (CAP) Platform is a modern, web-based application designed to empower fresh graduates and final-year students as they transition into the professional world. In an increasingly competitive job market, many students struggle with interview preparation, lack of industry exposure, and difficulty presenting their academic achievements in a professional format. CAP addresses these challenges by providing a unified space for interview preparation, portfolio creation, and seamless interaction with recruiters.

Unlike traditional resumes or scattered online profiles, CAP enables users to build a structured, visually appealing, and comprehensive digital academic portfolio that highlights their skills, projects, and accomplishments. The platform also integrates an AI Mock Interview Module, allowing students to experience simulated interview scenarios. This module evaluates communication clarity, confidence, and understanding of subject fundamentals, offering personalized feedback to help students improve their interview performance [1].

Developed using the Django web framework, PostgreSQL, and following the Agile development methodology, CAP ensures a user-friendly and secure environment accessible to individuals from all technical backgrounds. While the platform uses AI to support mock interviews and feedback, it intentionally avoids automated ranking, scoring, or biased decision systems ensuring a fair, transparent, and student-centered experience.

For recruiters, CAP provides a dedicated portal to explore candidate portfolios, apply filters based on skills or academic background, and post internships or entry-level job openings. Recruiters can directly connect with potential candidates without relying on automated fit scores, preserving human judgment and fostering meaningful engagement.

Security and reliability are ensured through authentication mechanisms, structured data handling, and thorough testing at every stage of development including unit testing, integration testing, system testing, and user acceptance testing.

This project, developed as part of the Bachelor of Science in Computer Science (2022–2026) program at COMSATS University Islamabad, Sahiwal Campus, is designed to be deployment-ready and capable of evolving through real-user feedback. CAP aims to create a digital ecosystem that strengthens student visibility, enhances interview readiness, and builds a direct bridge between academic achievement and early-career opportunities.

## **1.1 Brief Overview**

The Customized Academic Portfolio project aims to develop a web-based academic portfolio platform designed exclusively for students, researchers, and professionals to enhance how they manage, present, and grow their academic profiles. The system will be developed using HTML5, CSS3, JavaScript, Bootstrap 5, and jQuery for the front end; Python 3.12, Django 5.0.4, and Django REST Framework for the backend; and SQLite for development with PostgreSQL in production to ensure a secure, scalable, and high-performance experience. The project adopts agile software development methodology, enabling iterative and structured development [2].

Customized Academic Portfolio is a unique and focused platform designed to provide users with an intelligent space where they can build academic portfolios, track achievements, and receive AI-powered insights. While many professional tools exist, Customized Academic Portfolio stands out due to its specialized focus on academic growth, making it a dedicated hub for learners and researchers. It addresses challenges like scattered records, limited visibility, and weak career guidance by enabling users to organize their academic data, leverage AI, and access personalized recommendations ensuring they stay informed and prepared for opportunities that align with their academic journey.

## **1.2 Relevance to Course Modules**

The development of Customized Academic Portfolio directly applies and integrates various concepts and skills acquired throughout our Computer Science course.

### **i. Technical Skills:**

The project involves the application of multiple programming languages, frameworks, and tools learned during the course, including HTML5, CSS3, JavaScript, Bootstrap 5, jQuery, and backend development using Python and Django 5. These skills were essential for building the portfolio management system and implementing core functionalities such as user authentication, portfolio creation, publication tracking, and recruiter search. The principles of software engineering such as modular design, MVC architecture, and secure coding practices were applied to ensure system reliability, scalability, and maintainability.

### **ii. Software Engineering Principles:**

The project adheres to software engineering principles covering requirement analysis, system design, software architecture, testing, and agile development. Each phase of the Customized Academic Portfolio development follows structured practices to ensure scalability,



maintainability, usability, and security. The system integrates concepts from web development, distributed systems, version control, and cloud deployment making it a comprehensive computer science application.

### **iii. Database Design:**

A strong understanding of database management was essential in designing the SQLite (development) and PostgreSQL (production) databases for storing user profiles, academic achievements, projects, publications, skills, and recruiter search filters. The platform makes use of JSONB fields, relational mapping, indexing, and PostgreSQL full-text search to ensure fast and efficient retrieval of academic data across large datasets.

### **iv. System Architecture:**

The project required an understanding of modern web system architecture and cloud-based deployments. Concepts such as Django's MVT architecture, REST API design, microservices-ready structure, and containerization were applied to build a structured and scalable system. Deployment tools such as Docker, Kubernetes, and CI/CD pipelines ensure that the platform can scale efficiently and maintain high availability.

### **v. UI/UX Design:**

A user-friendly interface is essential for the success of an academic platform. Concepts from human-computer interaction and UI/UX design courses guided the creation of clean, responsive layouts using Bootstrap 5 and Tailwind CSS. These principles helped build an intuitive and visually consistent interface that students, researchers, and recruiters can use comfortably across devices.

### **vi. Project Management:**

Project management concepts played a critical role in planning, executing, and controlling the development phases of Customized Academic Portfolio. Agile methodology supported iterative development, sprint-based progress, and continuous improvement. This ensured that features such as portfolio modules, recruiter dashboards, and AI-based recommendations were delivered in a timely and structured manner.

### **vii. Networking and AI Integration:**

Customized Academic Portfolio includes AI-driven features such as skill extraction, smart recommendations, candidate ranking, and academic pattern analysis, which rely on concepts from machine learning and NLP using tools like spaCy, BERT, TensorFlow, and scikit-learn. Additionally, real-time updates such as recruiter portfolio edits, and search filtering depend on efficient API communication and backend networking principles. Understanding HTTP

protocols, REST standards, and asynchronous operations ensures seamless interaction across the platform.

### **1.3 Project Background**

The Customized Academic Portfolio (CAP) Platform was conceptualized in October 2024 at COMSATS University Islamabad, Sahiwal Campus, under the supervision of Mr. Ahmad Shaf, as part of the Bachelor of Science in Computer Science (2022–2026) final year project. The idea originated from the real difficulties faced by fresh graduates and final-year students during their transition into the professional world. Many students shared that they felt unprepared for job interviews, especially when experiencing them for the first time, and struggled to understand industry expectations. Recruiters similarly expressed that evaluating fresh graduates was often challenging due to incomplete CVs, inconsistent presentation of academic work, and the absence of a unified platform showcasing a student's capabilities.

A survey conducted with forty participants, including students, faculty members, and recruiters, revealed widespread issues in early-career preparation. Students reported a lack of confidence in interviews, difficulty explaining their academic projects, and confusion about how to build a professional identity beyond a basic CV. Many highlighted that traditional profiles such as static resumes or simple LinkedIn descriptions failed to represent their skills, academic achievements, and potential in a structured and compelling manner. Recruiters, on the other hand, noted that fragmented information across multiple platforms made it difficult to gain a clear understanding of a candidate's overall readiness.

The CAP Platform was designed to address these concerns by providing a dedicated space where students can create a well-organized and visually appealing academic portfolio, practice interviews through an AI-assisted mock interview module and interact directly with recruiters without relying on automated ranking or algorithmic scoring. The mock interview system helps students experience realistic interview scenarios and receive constructive feedback, while the portfolio builder allows them to present their academic projects, skills, and achievements in a refined and professional format. Recruiters benefit from an accessible and streamlined interface where they can explore student portfolios, post entry-level job or internship openings, and connect with potential candidates based on their profiles rather than algorithmic recommendations.

Developed using the Django web framework, PostgreSQL, and Agile methodology, the project emphasizes secure authentication, efficient data management, and an intuitive user experience suitable for individuals from all technical backgrounds. Over a six-month development

timeline, the platform progressed through requirement analysis, design, implementation, testing, and deployment phases, ensuring reliability and usability. CAP aligns with the university’s mission to encourage innovation and practical skill development, offering a scalable solution that enhances student visibility, strengthens interview readiness, and facilitates meaningful engagement between graduates and recruiters.

## 1.4 Literature Review

Table 1.1 states that Customized Academic Portfolio platform builds upon foundational and contemporary research in Natural Language Processing (NLP), recommendation systems, web-based portfolio management, and academic profiling. This review synthesizes key studies and frameworks that validate the need for an AI-integrated, intelligent academic portfolio system as given in table 1.1.

**Table 1.1** Literature Review

| <b>Application Name</b>                                   | <b>Weakness</b>                                                                              | <b>Proposed Project Solution (Customized Academic Portfolio)</b>                                                                |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Academic Portfolio Platforms</b>                       | Limited customization, basic templates, lack of AI insights, scattered academic data.        | Fully customizable academic portfolios, AI-powered skill extraction, publication management, and structured academic profiling. |
| <b>Professional Networking Platforms (e.g., LinkedIn)</b> | Not tailored for students/researchers, mixed professional focus, limited academic analytics. | Academic-focused environment with research-oriented profiles, achievements tracking, and AI-based academic insights.            |
| <b>Online CV/Resume Builders</b>                          | Static templates, no real-time updates, no integration with academic data sources [3].       | Dynamic portfolio builder with live updates, academic sections, publication syncing, and                                        |

|                                             |                                                                                               |                                                                                                                             |
|---------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                             |                                                                                               | recruiter-ready export options.                                                                                             |
| <b>University Management or LMS Systems</b> | Focus on academics only; missing career tools, portfolio features, and researcher visibility. | Dedicated space for showcasing skills, research work, certifications, projects, and academic growth to external recruiters. |
| <b>Job Portals / Recruitment Platforms</b>  | Generic candidate ranking, lack of academic depth, no skill-gap analysis [4].                 | Smart recruiter dashboard with AI-based candidate matching, ranking, skill-gap analysis, and advanced search with filters.  |

## 1.5 Analysis from Literature Review

This section evaluates findings from the literature related to digital portfolios, interview preparation, early-career recruitment challenges, and UI/UX best practices. The reviewed research emphasizes the importance of structured portfolio presentation, student preparedness for interviews, and platforms designed specifically for fresh graduates. Studies highlight that students benefit most from systems that offer clarity, guided self-presentation, and opportunities for recruiter visibility. Research also underscores the limitations of traditional CVs and scattered online profiles, suggesting the need for centralized platforms that support academic identity building without overwhelming technical complexity.

Unlike platforms relying on automated ranking or algorithmic scoring, the literature supports solutions that enhance user confidence, improve communication skills, and promote transparent recruiter–candidate engagement. These insights justify CAP’s focus on mock interview assistance, portfolio structuring, and direct hiring pathways, ensuring students gain both confidence and visibility as they transition into professional environments.

### 1.5.1 Academic Portfolio Platforms

Existing academic portfolio platforms generally offer static templates and minimal personalization. Many lack the ability to comprehensively represent a student's academic work, projects, and skills in an organized and professional manner. CAP addresses this gap by

enabling users to build detailed, visually appealing portfolios tailored to their field of study and early-career goals. The platform emphasizes clarity, structured presentation, and user-friendly editing, allowing students to present their academic identity more effectively.

### **1.5.2 Professional Networking Platforms**

Professional networking platforms such as LinkedIn serve broad professional audiences, mixing corporate achievements with academic experiences. They are not specifically designed for students entering the job market for the first time, and they lack structured sections for academic projects, coursework, and early-stage accomplishments. CAP resolves this by offering an academic-focused environment dedicated to showcasing a student's learning journey, technical work, and career preparation efforts without distraction from unrelated content.

### **1.5.3 Online CV/Resume Builders**

Traditional CV builders provide static, one-page templates that lack depth and do not help students articulate their academic strengths or prepare for interviews. They also do not integrate mock interviewing or guidance features, leaving fresh graduates unsure about how to communicate their skills effectively. CAP enhances this space by offering dynamic portfolio creation tools alongside an AI-assisted mock interview module, helping students practice responses, build confidence, and present their academic profiles professionally.

### **1.5.4 University Management or LMS Systems**

Although LMS platforms provide access to course materials and academic services, they do not include tools for showcasing academic achievements, creating portfolios, or improving career visibility. Customized Academic Portfolio fills this gap by enabling students to present their skills, certifications, projects, and research work in a structured manner while offering AI-driven suggestions to improve academic and professional readiness.

### **1.5.5 Job Portals and Recruitment Platforms**

Traditional job portals offer generic candidate matching without a deep understanding of academic work, research output, or skill progression. They often prioritize experience over academic potential. Customized Academic Portfolio introduces an advanced recruiter dashboard with AI-based candidate ranking, skill-gap analysis, fine-grained filtering, and personalized profile recommendations, making academic talent more visible and accessible to recruiters.

## 1.6 Methodology and Software Lifecycle

Customized Academic Portfolio is a software project with diverse and evolving requirements, making it important to use a development methodology that supports flexibility, collaboration, and continuous refinement. Agile is an ideal fit because it allows the system to grow iteratively while adapting to user needs across students, researchers, and recruiters.

### 1.6.1 Rationale Behind Selected Agile Model

Agile emphasizes adaptability, teamwork, and ongoing enhancement, making it well suited for a platform like Customized Academic Portfolio that requires frequent updates and user-driven improvements. Since the system involves continuous interaction with users such as refining portfolios, updating academic data, and improving AI-based recommendations, the Agile model ensures the platform evolves in sync with real-world academic and professional needs.

Agile is based on the following principles:

1. **Iterative development:** Core features such as portfolio creation, academic records, AI insights, and recruiter tools are developed in cycles, with each iteration refining functionality.
2. **Incremental delivery:** Modules are released in small, usable increments, enabling early access to features like dashboards, profile components, and search filters.
3. **Continuous feedback:** Students, researchers, and recruiters provide ongoing input regarding usability, academic relevance, and recommendation accuracy.
4. **Adaptive planning:** The development team refines plans based on feedback, evolving academic workflows, and AI model behavior, ensuring CAP remains aligned with user expectations.

## 1.7 Summary

The Customized Academic Portfolio (CAP) Platform, initiated in October 2024 at COMSATS University Islamabad, Sahiwal Campus, was developed to address the practical challenges faced by fresh graduates, final-year students, and recruiters in presenting and evaluating academic and early-stage professional achievements. Traditional methods such as static CVs, scattered online profiles, and manually created portfolios often fail to provide a structured, complete, and professional representation of a student's work. Existing platforms like LinkedIn, Google Scholar, Academia.edu, and online resume builders either lack academic depth, mix corporate and academic content, or require technical design skills that many students do not possess.

To overcome these limitations, the Customized Academic Portfolio platform provides a centralized, user-friendly system where students can build structured academic portfolios, highlight projects, showcase skills, and present achievements in a polished and professional manner. In addition, CAP introduces an AI-assisted mock interview feature that enables students to practice interview scenarios, receive constructive feedback, and improve their communication and confidence before appearing in real interviews. For recruiters, the platform offers a dedicated interface that allows them to browse student portfolios, apply filters related to degree or skills, and connect directly with potential candidates without relying on automated ranking or algorithmic scoring.

Developed using the Agile methodology, CAP emphasizes iterative development, continuous feedback, and adaptive planning to ensure that the platform meets real user needs. Through modern web technologies, secure authentication, and an accessible interface, the system supports smooth portfolio creation, reliable data handling, and an environment that enhances student visibility and improves early-career readiness. CAP ultimately serves as a comprehensive digital space for academic representation, interview preparation, and meaningful recruiter–candidate interaction.

## **Chapter 2**

# **Problem Definition**



## **2 Problem Definition**

This chapter discusses the precise problem to be solved and the intended outcome.

### **2.1 Problem Statement**

Students and fresh graduates often struggle to organize, present, and showcase their academic achievements due to the absence of a centralized and dedicated platform. Traditional methods such as static resumes, scattered documents, manually updated CVs, and general professional networking sites fail to provide an academic-focused environment. As a result, students face difficulties highlighting their projects, skills, certifications, and accomplishments in a structured manner, leading to incomplete representation of their academic journey and reduced visibility when pursuing internships, research roles, or entry-level job opportunities. Recruiters also find it challenging to assess fresh candidates when important academic information is fragmented across multiple unrelated platforms.

To address these challenges, the Customized Academic Portfolio (CAP) platform is being developed as a unified academic and early-career preparation system. CAP enables students to create well-organized, visually appealing academic portfolios that clearly represent their skills, projects, and achievements. The platform also includes an AI-assisted mock interview feature, allowing users to practice interview scenarios, improve confidence, and better prepare for real hiring processes [5]. For recruiters and institutions, CAP provides a dedicated dashboard where they can explore student portfolios, filter candidates based on basic criteria such as skills or degree programs and directly connect with potential hires.

### **2.2 Deliverable and Development Requirements**

The Customized Academic Portfolio (CAP) project's deliverables and development requirements encompass a range of components essential for the successful implementation of an intelligent academic portfolio and recommendation platform. These components are detailed below:

#### **2.2.1 User Interface (UI) Design**

Develop a clean, intuitive, and academically focused interface that allows users to easily create, edit, and view their academic portfolios. The interface should be visually appealing and consistent across all modules, ensuring users can navigate without confusion. Responsiveness across devices, accessibility compliance, and clear visual hierarchy will be prioritized to

enhance user engagement and satisfaction. Design elements should also accommodate multilingual support in future expansions, making the platform inclusive for diverse academic communities.

### **2.2.2 User Registration and Profiles**

Implement a secure registration and login system where students, researchers, and professionals can create customizable academic profiles. Profiles will include sections for education, skills, projects, research papers, certifications, and achievements. Role-based access control will ensure appropriate permissions for different user types. The system should also allow users to update information dynamically, maintain historical records of portfolio changes, and integrate seamlessly with AI-driven recommendation engines.

### **2.2.3 Portfolio Builder**

Provide a structured and dynamic portfolio-building module where users can upload documents, add academic data, manage publications, and generate professional portfolios in multiple formats. The module should support drag-and-drop functionality, template customization, and real-time preview of portfolios. Users should also be able to categorize entries, tag skills, and maintain consistency in formatting. The builder must integrate with backend services to allow instant AI-based analysis and recommendations based on portfolio content.

### **2.2.4 AI-Powered Insights and Recommendations**

Integrate NLP and machine learning models to generate skill analysis, personalized recommendations, automated tagging of academic content, and smart suggestions for improving profiles. The AI engine should learn from user behavior, portfolio updates, and interaction history to refine insights over time. Recommendations must be contextual, actionable, and presented clearly to users. Additionally, the system should maintain logs for AI decisions to support transparency and continuous improvement.

### **2.2.5 Recruiter Dashboard**

Develop an advanced recruiter module featuring AI-based candidate ranking, skill-gap analysis, advanced search filters, and automated profile recommendations tailored to recruiter needs. The dashboard should provide analytics on candidate engagement, portfolio views, and application trends. Recruiters should be able to bookmark, shortlist, and export candidate lists

for offline evaluation. Integration with messaging and notification modules will ensure smooth communication between recruiters and candidates.

### **2.2.6 Search and Discovery**

Build efficient search functionality using PostgreSQL full-text search and JSONB fields, enabling users and recruiters to search for profiles, skills, research areas, and academic contributions with high accuracy. Search results should be filterable, sortable, and ranked according to relevance and AI-driven scoring. Autosuggestions and search history features will enhance user experience. The system must handle large datasets efficiently and provide fast response times even under high load.

### **2.2.7 Security and Privacy**

Ensure strong data protection using secure authentication, encrypted data handling, access control mechanisms, and adherence to modern security standards to protect academic information. Regular security audits and compliance with GDPR-like principles will be maintained. The platform should implement secure session management, multi-factor authentication, and safeguards against common threats such as SQL injection or cross-site scripting (XSS).

### **2.2.8 Scalability**

Architect the platform for scalability by leveraging Django REST Framework, PostgreSQL, Docker, and Kubernetes, ensuring the system can support increasing data volumes, users, and recruiter activity. Horizontal and vertical scaling strategies should be planned to maintain consistent performance. The database schema and backend services must support sharding, indexing, and caching mechanisms for efficient data retrieval.

### **2.2.9 Web Application Development**

Since Customized Academic Portfolio is a web-based system, the platform will be developed using HTML5, CSS3, JavaScript, Bootstrap 5, Tailwind CSS, and jQuery for the frontend, with Django and DRF powering the backend. Frontend components must be modular, reusable, and responsive. Backend APIs should follow RESTful principles, be well-documented, and allow seamless integration with AI modules and third-party services.

### **2.2.10 Documentation**

Prepare comprehensive user documentation, API documentation, and developer guides to ensure smooth onboarding, maintenance, and future system enhancements. Documentation

should include step-by-step instructions, screenshots, and code examples where applicable. Maintaining updated documentation will support knowledge transfer, facilitate debugging, and accelerate new feature integration.

#### **2.2.11 Testing and Quality Assurance**

Conduct rigorous testing, including unit tests, integration tests, and system tests to ensure the platform is stable, efficient, and error-free. Automated testing frameworks should complement manual testing to cover both edge cases and typical workflows. Test reports should document functionality, performance, security, and usability findings, providing actionable insights for continuous improvement.

#### **2.2.12 User Training and Support**

Develop user guides, tutorial videos, and support channels to help students and recruiters effectively use the platform and understand its AI-driven features. Training resources should cover both basic operations and advanced functionalities. Support channels, including FAQs, chatbots, and helpdesk services, will assist users in resolving issues promptly and maximizing platform adoption.

#### **2.2.13 Regular Updates and Maintenance**

Establish an ongoing maintenance and upgrade plan to introduce new academic templates, refine AI models, improve performance, and ensure continued system security. This plan will include periodic system audits, usability testing, and performance benchmarking to identify areas for improvement. Regular updates will ensure compatibility with evolving technologies, incorporate user feedback, and maintain a seamless, high-quality user experience.

These deliverables and development requirements collectively ensure the successful development and long-term evolution of the Customized Academic Portfolio platform. They align closely with the project's vision of supporting academic growth, enhancing professional visibility, and enabling smarter recruiter–candidate matching. Furthermore, they establish a structured foundation for integrating future AI-driven features, improving user experience, and scaling the system as the user base expands. By adhering to these requirements, the platform maintains consistency, reliability, and adaptability throughout its lifecycle.

### **2.3 Current System Limitations**

Currently, there is no centralized platform that enables students and fresh graduates to professionally create, manage, and showcase their academic achievements in a structured and

visually appealing way [6]. Most individuals rely on static resumes, scattered documents, and general professional networking sites, which are not tailored to academic presentations [7]. As a result, students often struggle to highlight their projects, skills, certifications, and accomplishments in a cohesive format, leading to incomplete representation and reduced visibility when applying for internships, research roles, or entry-level jobs [8].

Users also lack guided tools that help them prepare for interviews, build confidence, and understand how to communicate their academic work effectively [9]. Without a dedicated system to practice interviews or develop a structured academic portfolio, many students experience confusion, low confidence, and difficulty presenting themselves to recruiters.

The following table 2.1 summarizes the limitations of existing academic and professional platforms, emphasizing the need for an integrated, user-friendly solution like the Customized Academic Portfolio (CAP), which focuses on portfolio creation, interview preparation, and direct recruiter–student engagement. These gaps highlight the absence of a unified system that streamlines student development workflows while supporting personalized career growth. The analysis further reinforces the importance of a platform that bridges academic achievements with real-world recruitment opportunities as given in table 2.1.

**Table 2.1** Current System Limitations

| <b>Platform</b>                  | <b>Limitations</b>                                                                                                                                                                  |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LinkedIn</b>                  | Not designed for students or fresh graduates; lacks structured academic templates; mixes corporate and academic content; limited support for showcasing detailed academic projects. |
| <b>Academia.edu</b>              | Focuses mainly on research sharing; lacks a portfolio builder, skill presentation features, and dedicated tools for early-career job seekers or recruiters.                         |
| <b>Google Sites / Wix</b>        | Requires design and layout skills; no guided portfolio structure; no built-in interview preparation or recruiter interaction features [10].                                         |
| <b>Traditional CVs (PDF/DOC)</b> | Static and difficult to update; cannot display projects interactively; limited                                                                                                      |

|              |                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|              | visibility; lacks centralization and professional presentation for academic achievements [11].                                                    |
| <b>ORCID</b> | Primarily ID and publication tracking; not a visual portfolio platform; lacks sections for skills, projects, certifications, or recruiter access. |

## 2.4 Summary

The Customized Academic Portfolio (CAP) project addresses the challenges faced by students and fresh graduates in organizing and presenting their academic achievements by offering a centralized, structured, and user-friendly platform. Existing tools like LinkedIn, Academia.edu, and traditional CVs rely on static templates and scattered information, making it difficult for users to showcase skills, projects, and certifications or prepare for interviews. CAP overcomes these limitations by enabling users to create visually appealing portfolios, practice through an AI-assisted mock interview module, and connect directly with recruiters via a dedicated dashboard. Built using modern technologies such as Django, PostgreSQL, HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS, and jQuery, CAP ensures secure authentication, smooth navigation, and scalable performance.

# **Chapter 3**

## **Requirement Analysis**

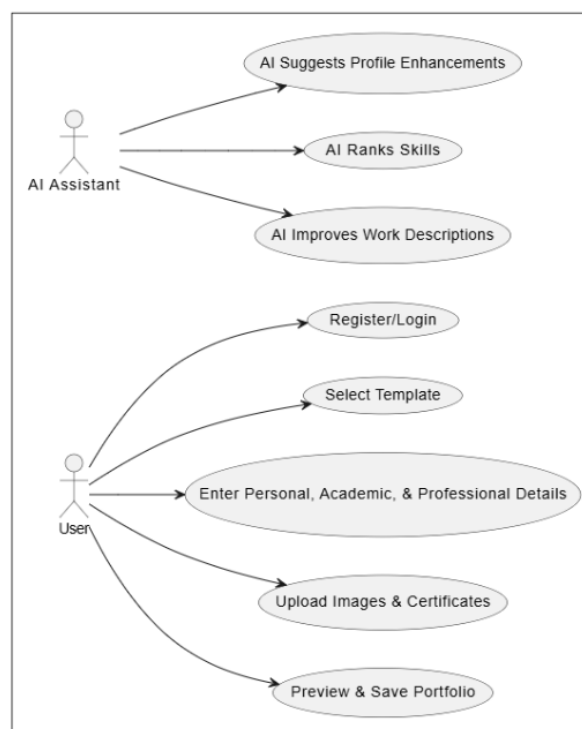
## 3 Requirement Analysis

### 3.1 Use Case Diagrams

In this Use case diagrams, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system [12].

#### 3.1.1 User Profile and Portfolio Management

Figure 3.1 illustrates the interactions between a User creating a portfolio and an AI Assistant enhancing it [13]. The User performs core creation tasks like Register/Login, Select Template, Enter Details, Upload, and Preview and Save. Concurrently, the AI Assistant provides value-added services by Suggesting Profile Enhancements, Ranking Skills, and Improving Work Descriptions.

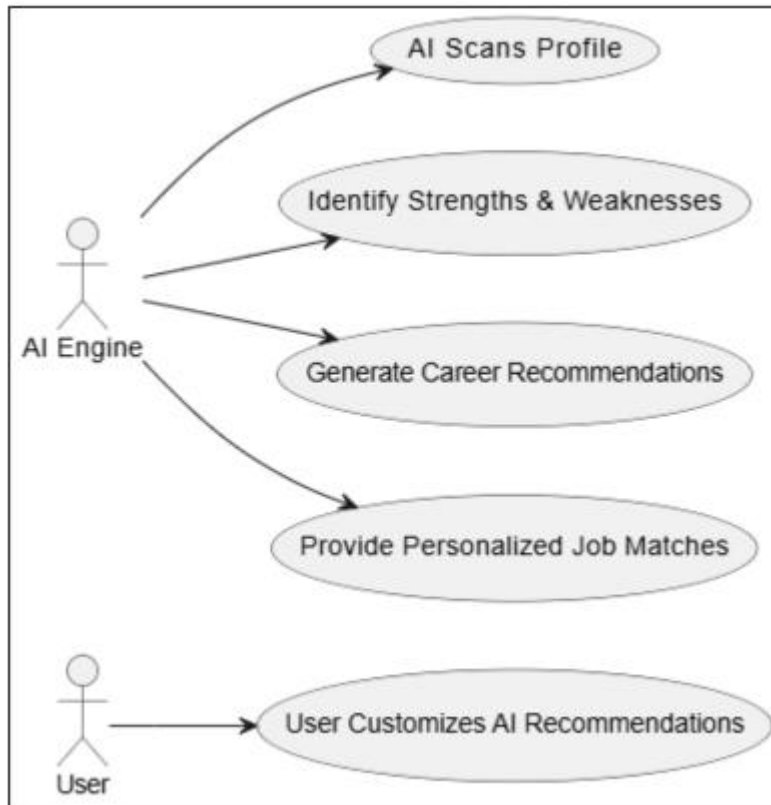


**Figure 3.1** User Profile and Portfolio Use Case Diagram

#### 3.1.2 AI Insights and Career Recommendations

Figure 3.2 outlines how the AI Engine analyzes a profile by Scanning it to Identify Strengths and Weaknesses, then using these insights to Generate Career Recommendations and Provide Personalized Job Matches, which the User can then Customize [14].



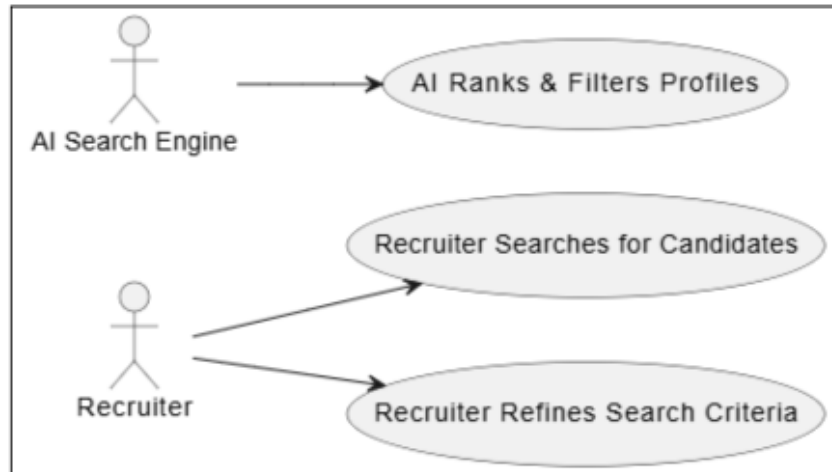


**Figure 3.2** AI Insights and Career Recommendations Use Case Diagram

### 3.1.3 Recruiter and Talent Search Dashboard

Figure 3.3 illustrates the end-to-end Talent Search workflow within the Customized Academic Portfolio platform [15]. It begins with the recruiter initiating a search for potential candidates by entering keywords, job titles, required skills, or experience levels. Once the initial search is performed, the recruiter can refine the results further through advanced filtering options such as education level, certifications, job roles, portfolio strength, skill proficiency, and location preferences. This interactive search-refinement loop helps recruiters narrow down a large pool of applicants into a smaller, more relevant shortlist that closely matches the hiring criteria.

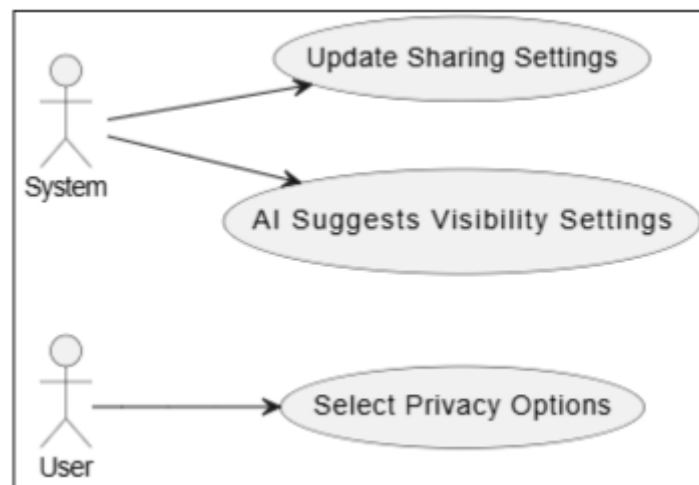
Parallel to the recruiter's manual search actions, the AI Search Engine continuously enhances the experience by intelligently analyzing candidate profiles. The AI system automatically ranks and filters profiles based on relevance, profile completeness, skill-match scores, and similarity to the job requirements. By leveraging machine learning and natural language processing, the system ensures that the most suitable candidates appear at the top of the results, reducing recruiter effort.



**Figure 3.3** Recruiter and Talent Search Dashboard Use Case Diagram

### 3.1.4 Sharing and Privacy Controls

Figure 3.4 illustrates how the User maintains full control over their profile by being able to Select Privacy Options according to their preferences [16]. The System supports this by allowing users to Update Sharing Settings, ensuring that only the intended information is visible to recruiters or the public. Alongside this, the AI assists the user by suggesting optimal Visibility Settings based on profile completeness, job-seeking status, and recruiter engagement trends. These intelligent recommendations help users decide which information should be highlighted to improve their chances of being discovered.



**Figure 3.4** Sharing and Privacy Controls Use Case Diagram

## 3.2 Use Case Description:

### 3.2.1 User Profile and Portfolio Management

Table 3.1 defines how users create, customize, and maintain their academic and professional portfolios within the Customized Academic Portfolio platform. It outlines the steps involved in profile setup, template selection, AI-assisted enhancements, and final publication of the portfolio. The description ensures clarity in user interactions, system responsibilities, and AI-supported recommendations, forming the foundation for personalized portfolio generation.

**Table 3.1** User Profile and Portfolio Management

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>    | UC-01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Use Case Name:</b>  | <b>User Profile and Portfolio Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Actors:</b>         | Primary: User<br>Secondary: AI Assistant (for recommendations and enhancements)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description:</b>    | Users create and manage their profiles, portfolios, skills, experiences, and academics in one integrated system. AI assists with suggestions, ranking, and portfolio enhancement.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Trigger:</b>        | User registers, updates their profile, or customizes their portfolio.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Preconditions:</b>  | PRE-1. (Precondition) User must be logged in.<br>PRE-2. The system should have templates and database storage available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Postconditions:</b> | POST-1. (Postcondition) User profile and portfolio are saved and updated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Normal Flow:</b>    | <ol style="list-style-type: none"><li>1. User initiates the sign-up process and enters personal details, including name, email, and bio.</li><li>2. The system displays available portfolio templates for the user to choose from.</li><li>3. User selects a template for portfolio creation.</li><li>4. The system loads the selected template and prompts the user to enter personal, academic, and professional details.</li><li>5. User inputs personal, academic, and professional details, including education, work experience, achievements and uploads relevant images, certificates, and project links.</li></ol> |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <ol style="list-style-type: none"> <li>6. The system fetches user publications from Google Scholar (if connected) and suggests adding them to the portfolio</li> <li>7. The system processes the input data and AI suggests improvements for profile completeness.</li> <li>8. The system ranks the entered skills based on industry demand using AI.</li> <li>9. The system enhances descriptions for projects and work experience, improving readability and impact.</li> <li>10. User previews the final portfolio.</li> <li>11. The system displays portfolio preview with all entered details.</li> <li>12. User confirms and clicks "Save" to publish or update the portfolio.</li> <li>13. The system saves the portfolio, updates the user's profile, and generates a shareable portfolio link.</li> </ol>                                                                                                                                                     |
| <b>Alternative Flows:</b> | <ol style="list-style-type: none"> <li>1.1 User edits an existing portfolio instead of creating a new one. <ol style="list-style-type: none"> <li>1. Return to step 6 of normal flow.</li> </ol> </li> <li>1.2 AI auto-fills missing data based on uploaded resumes or profiles. <ol style="list-style-type: none"> <li>1. Return to step 6 of normal flow.</li> </ol> </li> <li>1.3 No Publications Found in Google Scholar. <ol style="list-style-type: none"> <li>1. If the user chooses to manually add publications, then the system allows manual entry.</li> <li>2. If the user skips, then the system proceeds without adding publications.</li> <li>3. Return to step 6 of normal flow.</li> </ol> </li> <li>1.4 User Removes Synced Publications <ol style="list-style-type: none"> <li>1. If the user deletes an auto-synced publication, then the system removes it from the portfolio</li> <li>2. Return to step 6 of normal flow.</li> </ol> </li> </ol> |
| <b>Exceptions:</b>        | <ol style="list-style-type: none"> <li>1.0.E1 (Exception) If a required field is missing, the system prompts the user. <ol style="list-style-type: none"> <li>1. If a required field is missing, then the system prompts the user to complete it.</li> <li>2a. If the user provides the missing information, then the system proceeds with normal flow.</li> </ol> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p>2b. Else if the user cancels the process, then the system terminates the use case.</p> <p>1.1.E1 If AI fails to generate suggestions, a default template is used.</p> <p>1. If AI fails to generate suggestions, then the system provides a default template.</p> <p>2. Return to step 6 of normal flow.</p> <p>1.2.E1 If the system crashes, unsaved changes are lost.</p> <p>1. If the system crashes, then any unsaved changes are lost</p> <p>2a. If the user restarts the system, then they must re-enter lost data.</p> |
| <b>Business Rules</b> | <p>BR-1 (Business Rule) Users can create multiple portfolios.</p> <p>BR-2 Verified certifications improve AI recommendations.</p> <p>BR-3 Each work experience entry must include at least one achievement.</p> <p>BR-4 If a user links to their Google Scholar account, publications are auto-synced and suggested for portfolio inclusion.</p> <p>BR-5 Users can manually add or remove publications even after auto-syncing.</p>                                                                                              |
| <b>Assumptions:</b>   | <p>1. Users prefer pre-designed templates.</p> <p>2. AI will assist but not fully automate profile creation.</p> <p>3. Users may refine project details later.</p>                                                                                                                                                                                                                                                                                                                                                               |

### 3.2.2 AI Insights and Career Recommendations

Table 3.2 explains how the Customized Academic Portfolio AI Engine analyzes user data to generate meaningful career insights, identify skill gaps, and suggest relevant opportunities. It highlights the interaction between the user and the AI system as it processes academic, professional, and skill-based information. The description also outlines how personalized recommendations evolve based on industry trends and user behavior, ensuring continuous and actionable career guidance.

**Table 3.2** AI Insights and Career Recommendations

|                       |                                                       |
|-----------------------|-------------------------------------------------------|
| <b>Use Case ID:</b>   | UC-02                                                 |
| <b>Use Case Name:</b> | <b>AI-Powered Career Insights and Recommendations</b> |
| <b>Actors:</b>        | Primary: AI Engine                                    |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Secondary: User                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Description:</b>       | AI analyzes user profiles and suggests career growth strategies, skill gaps, and personalized job opportunities.                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Trigger:</b>           | AI runs periodic analysis or user requests insights.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Preconditions:</b>     | PRE-1. (Precondition) Profile completeness is at least 60%.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Postconditions:</b>    | POST-1. (Postcondition) AI provides actionable career insights.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Normal Flow:</b>       | <ol style="list-style-type: none"> <li>1. AI scans user profile.</li> <li>2. Identifies strengths and weaknesses.</li> <li>3. Generate career and job recommendations.</li> </ol>                                                                                                                                                                                                                                                                                                           |
| <b>Alternative Flows:</b> | <ol style="list-style-type: none"> <li>1.1 User customizes AI recommendations.</li> <li>1. If the user customizes AI-generated recommendations, then the system updates the recommendations based on user preferences.</li> <li>2. Return to step 1 of the normal flow.</li> </ol>                                                                                                                                                                                                          |
| <b>Exceptions:</b>        | <ol style="list-style-type: none"> <li>1.0.E1 (Exception) AI fails to generate insights.</li> <li>1. If AI fails to generate insights, then the system informs the user that recommendations are unavailable.</li> <li>2a. If the user requests a retry, then AI attempts analysis again.</li> <li>2b. Else if the user exits the insights page, then the system terminates the process.</li> <li>2. Return to step 1 of the normal flow if insights are successfully generated.</li> </ol> |
| <b>Business Rules</b>     | <p>BR-1 (Business Rule) AI updates insights based on industry trends.</p> <p>BR-2 AI recommendations are personalized based on user activity, skills, and job market demand.</p> <p>BR-3 AI prioritizes verified data sources for generating career recommendations.</p> <p>BR-4 Users can override AI-generated insights, but the system keeps a lot of manual modifications.</p>                                                                                                          |
| <b>Assumptions:</b>       | <ol style="list-style-type: none"> <li>1. AI data sources remain updated.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                        |

### 3.2.3 Recruiter and Talent Search Dashboard

Table 3.3 case outlines how recruiters interact with the Customized Academic Portfolio platform to discover suitable candidates using AI-powered search and ranking. It explains how the system processes recruiter-defined criteria, refines results through intelligent filtering, and

presents the most relevant profiles. The description also highlights how dynamic updates and privacy rules ensure an efficient, compliant, and accurate talent–matching experience.

**Table 3.3** Recruiter and Talent Search Dashboard

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>       | UC-03                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Use Case Name:</b>     | <b>Recruiter and Talent Search</b>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Actors:</b>            | Primary: Recruiter<br>Secondary: AI Search Engine                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Description:</b>       | Recruiters search for candidates based on AI-powered filtering and ranking.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Trigger:</b>           | Recruiters search for talent.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Preconditions:</b>     | PRE-1. (Precondition) Recruiter accounts are verified.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Postconditions:</b>    | POST-1. (Postcondition) Relevant profiles are displayed.<br>POST-2. AI ranks best matches.                                                                                                                                                                                                                                                                                                                                         |
| <b>Normal Flow:</b>       | 1. Recruiters enter search criteria.<br>2. AI refines results and ranks profiles.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Alternative Flows:</b> | 1.1 Recruiter refines filters for better results.<br>1. If the recruiter is unsatisfied with the search results, then the system allows modification of search criteria.<br>2. If the recruiter refines the filters, then AI reprocesses the search and updates the rankings.<br>3. Else if the recruiter cancels the refinement, then the system maintains the current search results.<br>4. Return to step 1 of the normal flow. |
| <b>Exceptions:</b>        | 1.0.E1 (Exception) No matching profiles found.<br>1. If no profiles match the recruiter’s search criteria, then the system informs the recruiter and suggests broadening the filters.<br>2a. If the recruiter modifies the criteria, then the system re-runs the search.<br>2b. Else if the recruiter exits the search, then the system terminates the search process.                                                             |
| <b>Business Rules</b>     | BR-1 (Business Rule) Only public profiles are searchable.<br>BR-2 AI prioritizes candidates with higher profile completeness and relevance.                                                                                                                                                                                                                                                                                        |

|                     |                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | BR-3 Search results update dynamically as recruiters refine filters.<br>BR-4 AI ensures that recruiter searches comply with platform privacy policies. |
| <b>Assumptions:</b> | 1. Recruiters will refine searches iteratively.                                                                                                        |

### 3.2.4 Sharing and Privacy Controls

Table 3.4 describes how users manage the visibility, accessibility, and sharing permissions of their academic portfolios within the Customized Academic Portfolio platform. It explains how the system updates privacy configurations, ensures secure access, and supports AI-based suggestions to help users choose optimal visibility settings. The description also highlights built-in safeguards to prevent unauthorized sharing and maintain data protection standards.

**Table 3.4** Sharing and Privacy Controls

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>       | UC-04                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Use Case Name:</b>     | <b>Profile Sharing and Privacy Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Actors:</b>            | Primary: <b>User</b><br>Secondary: <b>System</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description:</b>       | Users control visibility and sharing of their portfolio links.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Trigger:</b>           | User updates privacy settings.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Preconditions:</b>     | PRE-1. (Precondition) User is logged in.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Postconditions:</b>    | POST-1. (Postcondition) Updated sharing settings are applied.                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Normal Flow:</b>       | 1. User selects privacy options.<br>2. System updates permissions.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Alternative Flows:</b> | 1.1 AI suggests best visibility settings based on career goals.<br>1. If the user is unsure about privacy settings, then AI analyzes the user's career goals and suggests optimal visibility settings.<br>2. If the user accepts AI suggestions, then the system automatically applies to the recommended settings.<br>3. Else if the user customizes settings manually, then the system updates permission based on user input.<br>4. Return to step 1 of the normal flow. |



|                       |                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exceptions:</b>    | 1.0.E1 (Exception) Unauthorized sharing attempt.<br>1. If an unverified user attempts to enable public visibility, then the system denies the request and notifies the user.<br>2a. If the user verifies their account, then the system allows public sharing.<br>2b. Else if the user does not verify, then the system maintains the current privacy settings. |
| <b>Business Rules</b> | BR-1 (Business Rule) Only verified users can enable public visibility.<br>BR-2 AI recommendations are based on industry best practices and user preferences.<br>BR-3 Users can override AI-suggested privacy settings.<br>BR-4 System logs all privacy setting changes for security auditing.                                                                   |
| <b>Assumptions:</b>   | 1. Users will update settings periodically.                                                                                                                                                                                                                                                                                                                     |

### 3.3 Functional Requirements

Functional requirements are essential in software development because they clearly specify what the system must do to fulfill users' needs and expectations. These requirements define the exact features, operations, and behaviors the system should provide, such as portfolio creation, data processing, AI-based recommendations, and recruiter search tools. Their importance lies in establishing a shared understanding among developers, users, and stakeholders, ensuring that the system's capabilities are fully aligned with project goals and user expectations. By identifying these functions early in the development lifecycle, functional requirements help avoid misunderstandings, minimize costly revisions, improve planning accuracy, and support thorough testing and validation. Moreover, they act as a foundation for system design, implementation, and evaluation. Ultimately, functional requirements guide the overall development process and ensure that the final product delivers meaningful, reliable, and user-centered value.

The functional requirements of the Customized Academic Portfolio (CAP) platform encompass a wide range of features and capabilities focused on enabling users to efficiently create academic portfolios, gain AI-driven insights, and connect with relevant academic or career opportunities. These requirements are designed to support students, researchers, professionals, and recruiters through structured workflows and intelligent tools. Each function contributes to

improving visibility, discoverability, and engagement within the academic and professional ecosystem. The key functional requirements include the following:

### **3.3.1 User Authentication and Authorization**

The system shall provide a secure login and registration mechanism for students, researchers, professionals, and recruiters. It will implement role-based access control to ensure that each user type can only access features relevant to their role within the Customized Academic Portfolio platform. This separation of permission enhances security, prevents unauthorized access, and ensures a personalized user experience. Additionally, the authentication system will support password recovery and session management to improve usability and reliability.

### **3.3.2 Portfolio Management**

The platform allows users to create, update, and customize their academic portfolios in a flexible and structured manner. Users can manage sections such as education, research work, publications, projects, skills, awards, certifications, and academic activities. The system will support customizable templates, media uploads, and section reordering to reflect individual academic journeys. This feature ensures that users can present their achievements professionally while maintaining consistency and clarity across portfolios.

### **3.3.3 Publication and Academic Data Sync**

The system shall enable automatic or manual syncing of academic records from external platforms such as Google Scholar, ORCID, and ResearchGate in future development phases. It will provide tools for importing publications, extracting relevant metadata, identifying keywords, and auto-categorizing academic records. This reduces manual effort and ensures data accuracy and consistency. Additionally, users will be able to review and edit imported data to maintain control over their academic profiles.

### **3.3.4 AI-Powered Insights and Recommendations**

The platform shall integrate AI-driven features to help users enhance and optimize their academic profiles. These features include career insights, skill recommendations, research topic suggestions, keyword extraction, and automated profile improvement tips tailored to each user's portfolio. By analyzing portfolio content and trends, the AI system supports informed decision-making and continuous improvement. This functionality adds intelligent value by guiding users toward stronger academic and professional positioning.

### **3.3.5 Recruiter Module and Talent Discovery**

The system should provide recruiters with tools to search, filter, and discover suitable candidates based on skills, publications, education, research interests, and experience. Advanced filtering options will allow precise talent discovery, while features such as bookmarking profiles and saving candidate lists improve recruitment efficiency. Recruiters will also have access to engagement analytics to evaluate profile views and interactions. This module bridges the gap between academic talent and recruitment opportunities.

### **3.3.6 Messaging Between Recruiters and Candidates**

The platform should include a secure and professional in-app messaging system to enable direct communication between recruiters and candidates. Recruiters can initiate conversations, share job details, and exchange documents, while candidates can respond and discuss opportunities seamlessly. Messaging will include timestamps, read receipts, and a clean, user-friendly chat interface. This feature ensures timely communication without requiring external platforms.

### **3.3.7 User Profile Management**

The system shall allow all users to manage and update their personal information, academic details, preferences, and interests. Users can view synced publications, saved portfolios, recruiter interactions, and portfolio analytics within their profiles. This centralized management helps users maintain up-to-date information and track their academic or professional progress. It also enhances personalization across platform features.

### **3.3.8 Search and Discovery**

The platform shall implement powerful search and discovery functionality to help users find academic portfolios, publications, research interests, and skill-based profiles. It will support keyword-based search, sorting options, and advanced filters to deliver fast and accurate results. This feature improves navigation efficiency and content discoverability across the platform. As a result, users can quickly locate relevant information and connections.

### **3.3.9 Feedback and Activity Tracking**

The system shall allow users to submit feedback regarding platform features and overall usability. It will also track user engagement metrics such as profile views, recruiter visits, publication visibility, and portfolio performance analytics. These insights help users understand their impact and identify areas for improvement. Feedback and tracking collectively support continuous platform enhancement and user growth.

## **3.4 Nonfunctional Requirements**

Nonfunctional requirements define the quality attributes, constraints, and performance standards that determine how the Customized Academic Portfolio platform operates. These requirements focus on system behavior rather than specific functionalities, ensuring reliability, efficiency, and positive user experience. They play a critical role in maintaining scalability, security, and usability as the platform grows. The key nonfunctional requirements are outlined below:

### **3.4.1 Performance**

The platform shall ensure fast loading times across all core features, including portfolio pages, publication syncing, search queries, and recruiter dashboards. AI-driven processes such as recommendation generation, keyword extraction, and analytics must execute efficiently without noticeable delays. The system should be optimized to handle high traffic volumes and thousands of concurrent users without performance degradation. Consistent performance is essential to maintaining user trust and engagement.

### **3.4.2 Security**

The system shall safeguard user data through strong encryption, secure authentication mechanisms, and strict access control policies. Sensitive academic information, recruiter details, and uploaded documents must be securely stored and transmitted. The platform will follow data protection principles similar to GDPR to ensure user privacy and transparency. Regular security checks and validations will further reduce the risk of unauthorized access or data breaches.

### **3.4.3 Availability**

The platform should maintain high availability to ensure users can access services at all times. Users should be able to update portfolios, sync publications, and respond to recruiter interactions without interruption. Minimal downtime and reliable server infrastructure are critical to supporting global usage. This requirement ensures continuity of academic and recruitment activities.

### **3.4.4 Ease of Use**

The platform's interface should be intuitive and user-friendly for students, researchers, and recruiters with varying levels of technical expertise. Navigation must be simple, consistent, and well-structured, with clear labels and visually clean layouts. The system will support

responsive design for all devices and consider accessibility standards to accommodate diverse users. Future enhancements may include multilingual support to further broaden usability and reach.

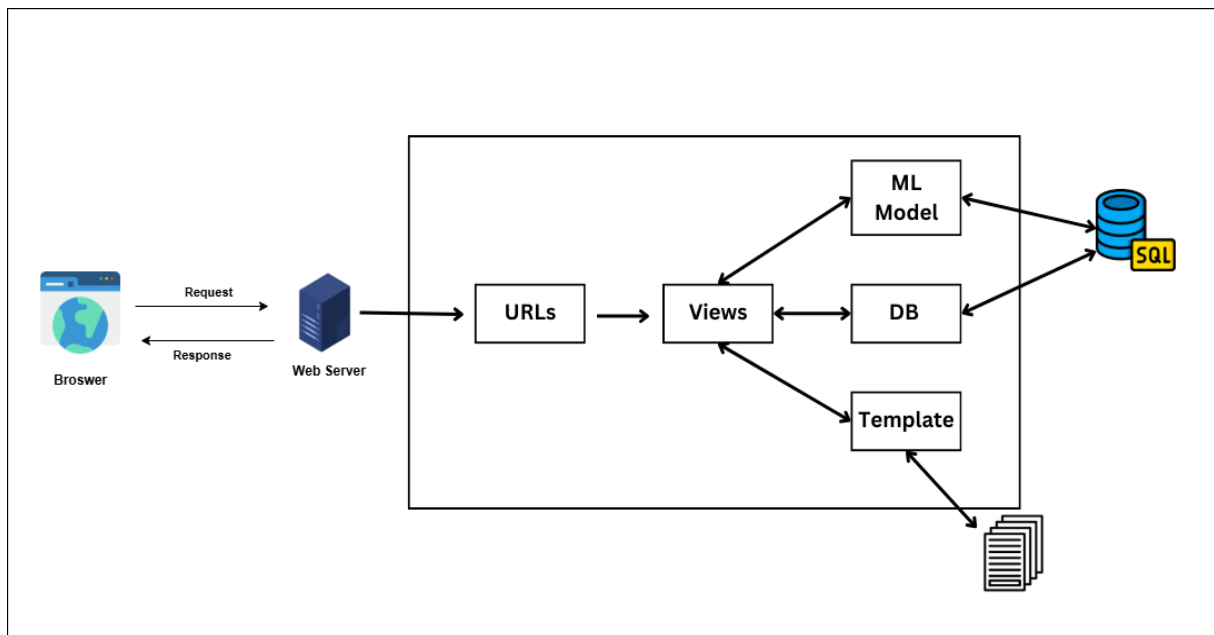
# **Chapter 4**

## **Design and Architecture**

## 4 Design and Architecture

### 4.1 System Architecture

Figure 4.1 explains how Customized Academic Portfolio operates by describing its core components, how they are organized, and the way they interact with each other to ensure smooth functionality throughout the platform.



**Figure 4.1** Customized Academic Portfolio Architecture

The architecture of the Customized Academic Portfolio (CAP) platform follows a modern client-server model designed to be modular, scalable, and easy to maintain. The client side is developed using HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS, and jQuery, providing fast, responsive, and intuitive interfaces for portfolio creation, profile management, mock interview interaction, and the recruiter dashboard. The backend is powered by Python 3.12, Django 5, and Django REST Framework, where the business logic manages core functionalities such as user authentication, portfolio management, mock interview handling, and recruiter-student interactions, ensuring a clean separation between the interface and application logic.

The system uses SQLite during development and PostgreSQL in production, offering reliable and secure storage for user profiles, academic portfolios, recruiter data, and mock interview records. The platform includes an AI-assisted mock interview module, which is limited to generating interview questions and providing practice-oriented interaction without any

ranking, scoring, or automated evaluation [17]. Role-Based Access Control (RBAC) ensures that students and recruiters can only access features relevant to their roles, while secure API communication and encrypted storage protect sensitive user data.

CAP's modular architecture supports future enhancements and scalability, allowing additional modules such as extended portfolio sections or recruiter tools to be integrated smoothly. This structure ensures efficient performance, maintainability, and a seamless user experience across all components of the platform.

## **4.2 Data Representation**

In the Customized Academic Portfolio (CAP) platform, all academic, portfolio, and recruiter-related data are stored within structured relational databases to ensure fast, consistent, and reliable access to information. The system utilizes PostgreSQL with well-defined relational schemas to systematically organize user profiles, publications, projects, skills, achievements, and recruiter search parameters while maintaining strong relationships among these entities. Unlike simple document-based or unstructured storage systems, this relational approach enables complex joints, indexing, and constraint enforcement, which are essential for maintaining data integrity and reducing redundancy.

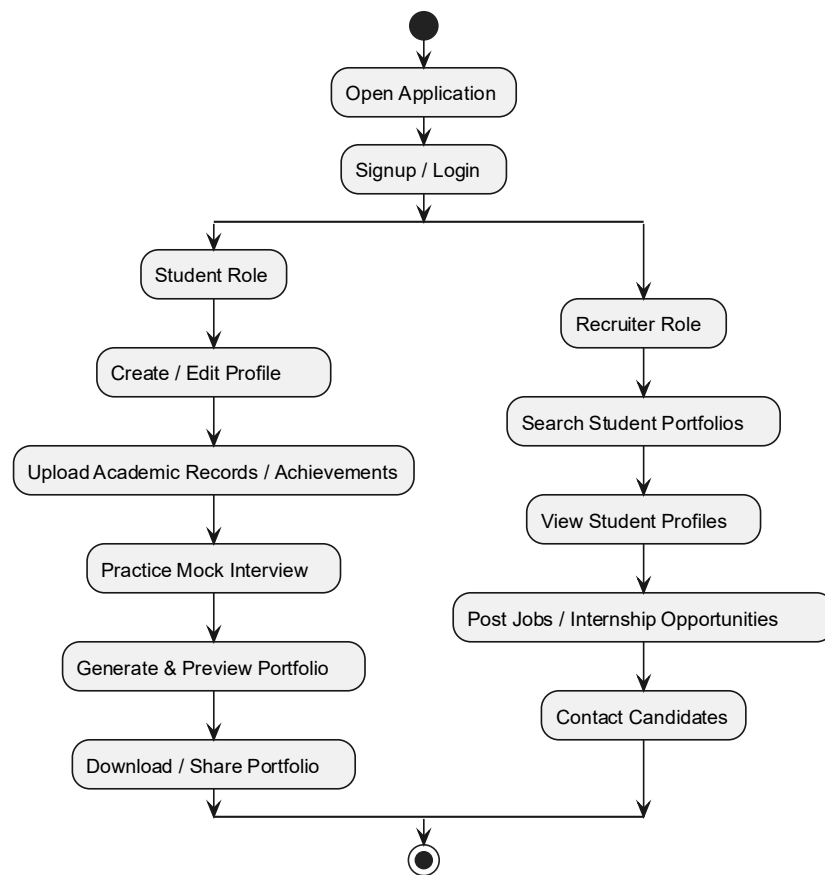
The structured data model allows CAP to support advanced querying mechanisms required for AI-based recommendations, keyword matching, profile ranking, and talent discovery. Efficient data encoding and decoding facilitate real-time updates, seamless synchronization of academic records, and responsive recruiter searches. Furthermore, the relational design ensures smooth integration with machine learning models by providing clean, normalized, and query-optimized datasets. As a result, this approach significantly enhances system accuracy, scalability, and performance, ultimately delivering a faster, more intelligent, and user-centric experience across platform [18].

## **4.3 Process Flow**

Figure 4.2 Process Flow Representation for the Customized Academic Portfolio (CAP) illustrates the complete journey of students and recruiters as they navigate the platform. It outlines how students register, build their academic portfolios, update their profiles, and practice through the AI-assisted mock interview module to improve confidence for real job interviews. The flow also shows how students manage their information, upload achievements, and maintain a structured, professional portfolio that can be viewed by potential employers.



Recruiters follow a parallel and simplified process where they register, authenticate, browse student portfolios, and post internships or job opportunities. They can directly view student profiles without ranking or scoring features, ensuring a transparent and straightforward recruitment experience. The diagram demonstrates how both user types interact smoothly with the system, highlighting an intuitive workflow and clear separation of roles. Overall, it captures how CAP brings efficiency, clarity, and seamless communication between fresh graduates and recruiters, ultimately creating a user-friendly environment that supports academic visibility and early-career hiring.



**Figure 4.2** CAP Process Flow Diagram

### 1. User Registration and Login

Users begin by creating an account and logging into the platform. After successful authentication, they are redirected to their personalized dashboard based on their role (student/researcher or recruiter).

### 2. For Students and Researchers

**Build Academic Portfolio:** Users can create and update their portfolio by adding education, skills, publications, projects, certifications, achievements, and experience.

**Upload Research Work:** They can upload research papers, articles, or external links, and sync publications from platforms like Google Scholar.

**Explore Opportunities:** Students can browse internship, scholarship, and research opportunities tailored to their academic background.

**AI Insights:** Users receive AI-driven recommendations for skills to improve, suitable opportunities, and portfolio enhancements [19].

**Manage Visibility:** They can control privacy settings and choose what information is visible to recruiters.

**Receive Notifications:** Students get alerts about profile views, recommended opportunities, AI suggestions, and recruiter messages.

### 3. For Recruiters

**Recruiter Dashboard:** Recruiters access a dedicated dashboard designed for talent search and candidate evaluation.

**Search Academic Profiles:** They can search and filter candidates based on skills, research areas, degrees, publications, and experience.

**View Detailed Portfolios:** Recruiters can view complete resource profiles, including research contributions, achievements, and skill sets.

**Bookmark and Save Talent:** Profiles can be saved for later review or added to custom talent pools.

**Contact Candidates:** Recruiters may send messages or interview requests directly through the platform.

**Receive Recommendations:** The system suggests candidates that match recruiter-defined criteria.

**Export Talent Data:** Recruiters can export selected candidate profiles for institutional use. This process ensures each user type receives appropriate tools and features tailored to their needs, enabling a seamless, intelligent, and structured experience within Customized Academic Portfolio.

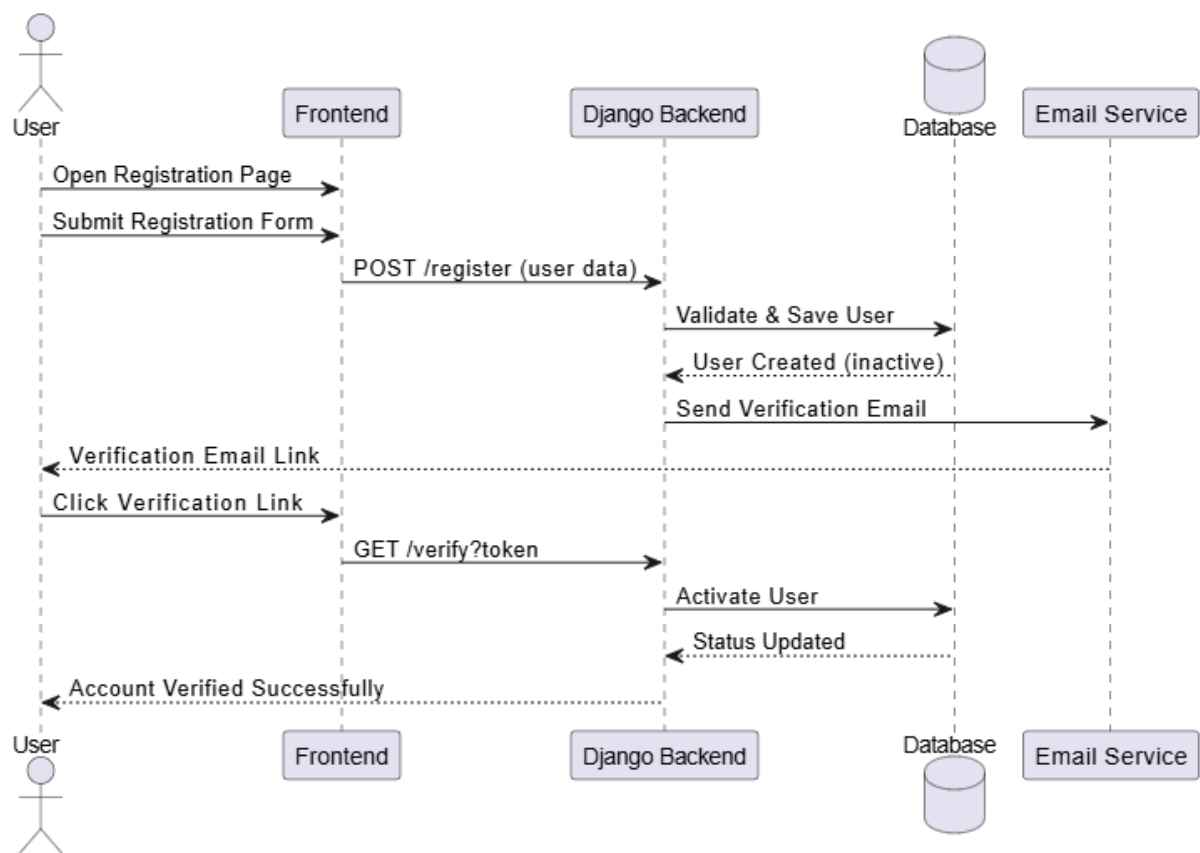
## 4.4 Sequence Diagram

### 4.4.1 User Registration and Email Verification

Figure 4.3 explains the complete registration flow in the CAP system. A new user fills out the HTML registration form, which Django validates and stores as an inactive account. The backend then sends a verification email to ensure that the user owns the email address provided.

Once the user clicks the verification link, a verification request is sent back to Django. The system validates the token, activates the account, and updates the database status. Finally, the user is notified that their account is successfully verified.

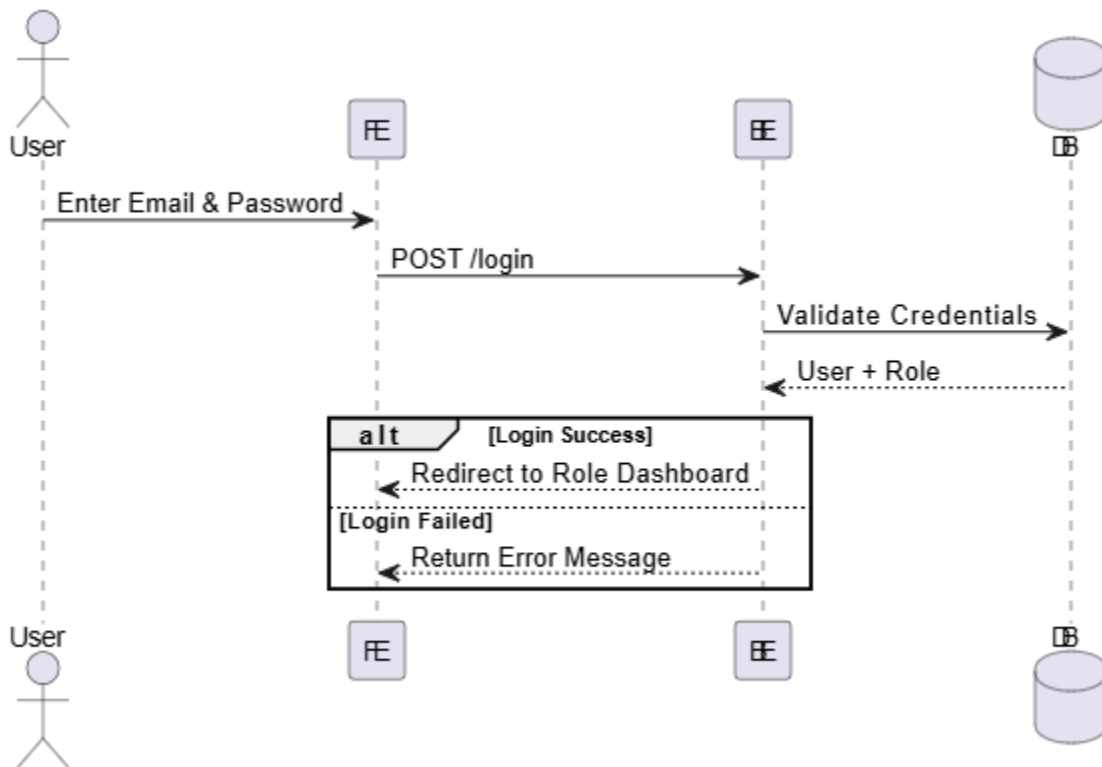
This process not only ensures secure and authenticated onboarding but also prevents unauthorized access and reduces the risk of fake or duplicate accounts. Additionally, the registration workflow is designed to handle errors gracefully, such as expired tokens or invalid email formats, providing clear feedback to users and maintaining a smooth onboarding experience.



**Figure 4.3** User Registration and Email Verification

#### 4.4.2 Login and Role-Based Dashboard

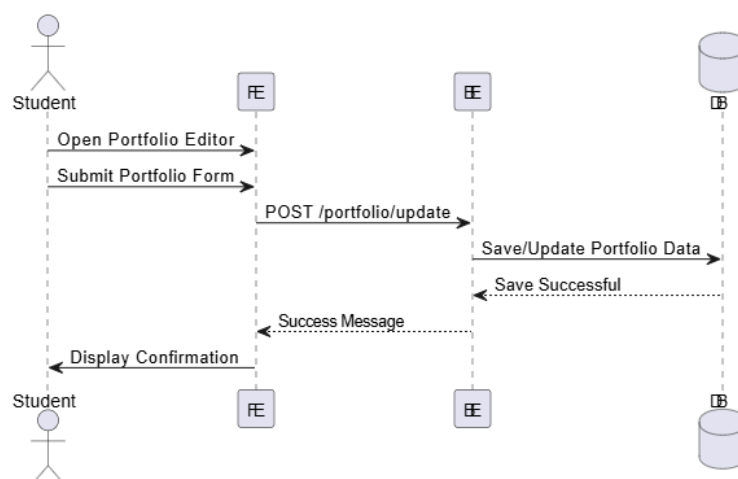
Figure 4.4 illustrates how users log into the CAP platform using a standard HTML login page. The entered credentials are submitted to Django, where authentication and role verification (student, recruiter, or admin) take place. If credentials are valid, Django redirects the user to the appropriate dashboard view based on their role. Incorrect credentials return an error message back to the frontend. Role-based routing ensures that each user accesses only the features intended for them. This maintains security and functionality separation.



**Figure 4.4** Login and Role-Based Dashboard

#### 4.4.3 Student Adds/Updates Portfolio

Figure 4.5 describes how a student manages their academic portfolio. The student interacts with HTML forms to add or update information such as skills, education, experience, and publications.

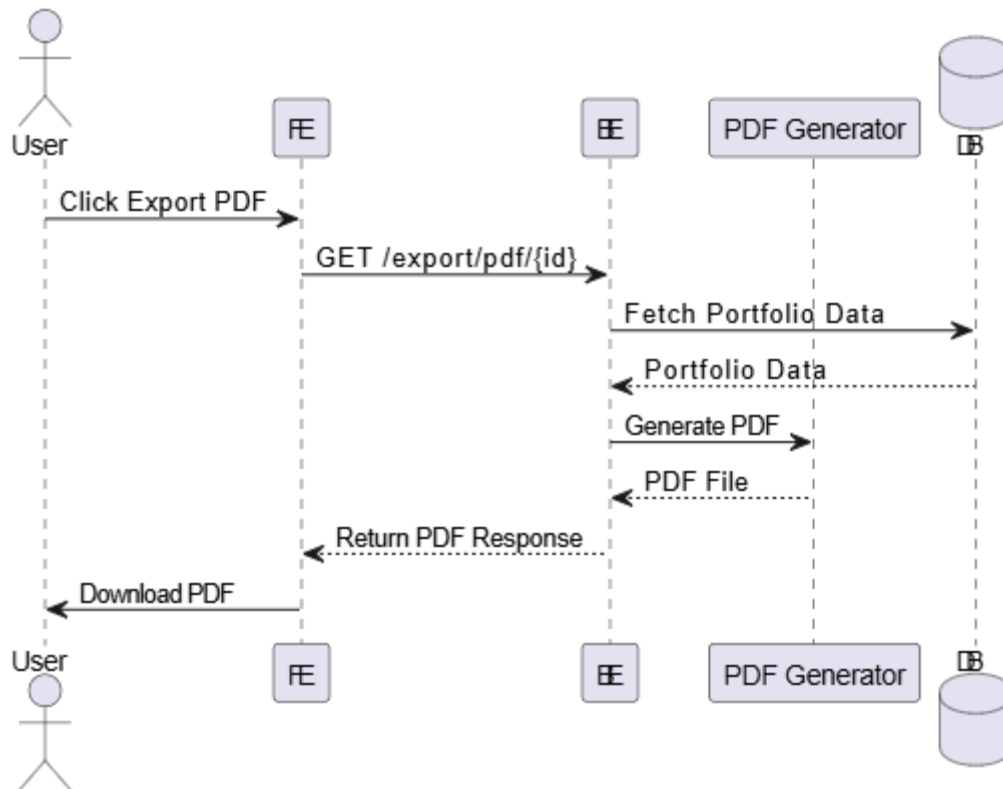


**Figure 4.5** Student Adds/Updates Portfolio

#### 4.4.4 Recruiter Searches Candidates

Figure 4.6 shows how recruiters search for suitable candidates using an HTML-based search interface. The recruiter applies filters such as skills, degree, location, or keywords. Django

receives these filters via GET request and performs database queries to retrieve matching students. The backend then returns a list of relevant candidates. The frontend displays the results clearly so recruiters can evaluate them. This search functionality enables fast talent discovery and reduces manual filtering. It significantly enhances recruitment efficiency in the CAP system.

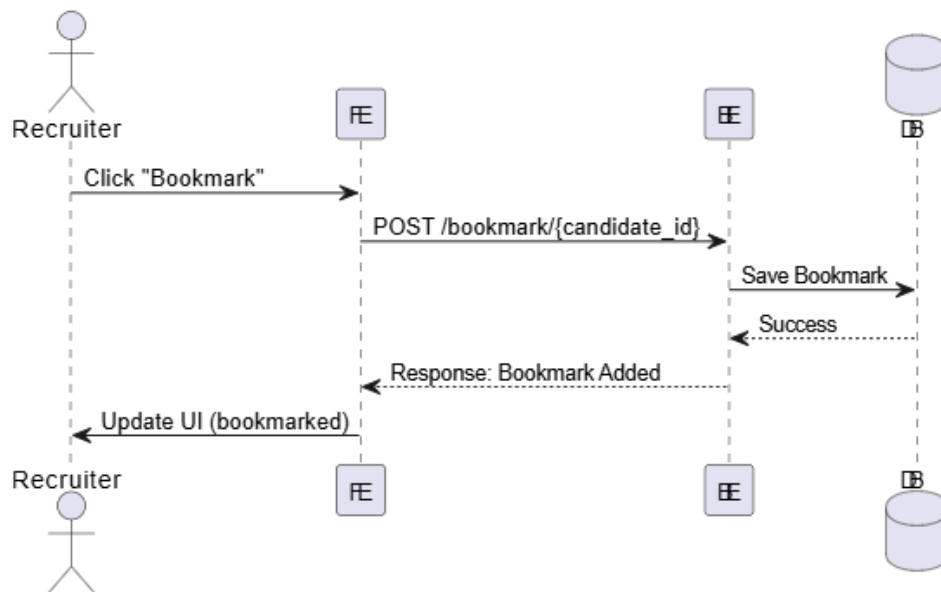


**Figure 4.6** Recruiter Searches Candidates

#### 4.4.5 Recruiter Views Candidate Portfolio

Figure 4.7 explains how recruiters access the detailed portfolio of a student. After selecting a candidate from search results, an HTTP request is sent to Django to retrieve the portfolio data. Django then fetches all associated sections, including projects, publications, achievements, and skills, from the database. The backend processes this information and renders a complete HTML page containing the candidate's academic and professional details. The recruiter can then view this well-organized portfolio for thorough assessment.

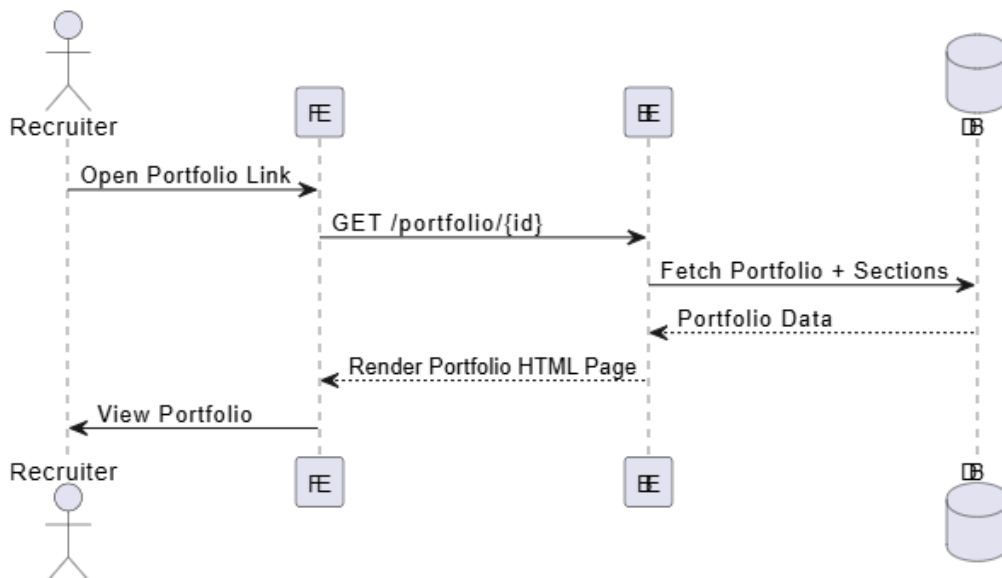
This centralized view not only improves decision-making and reduces evaluation time but also ensures consistency in data presentation across all profiles. It allows recruiters to perform detailed comparisons between candidates, identify key strengths, and make informed selection decisions efficiently.



**Figure 4.7** Recruiter Views Candidate Portfolio

#### 4.4.6 Bookmark Candidate

This diagram outlines how recruiters save candidates for later review using the bookmark feature. When the recruiter clicks the bookmark button, the front end sends a POST request to Django with the candidate’s ID. Django adds a bookmark entry into the database linked to that recruiter.

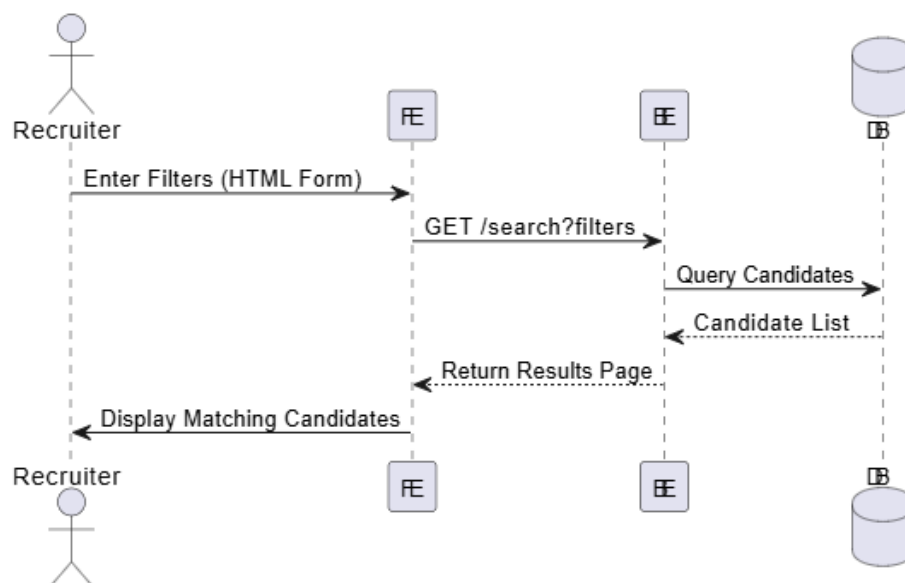


**Figure 4.8** Bookmark Candidate

#### 4.4.7 Export Portfolio as PDF

Figure 4.9 illustrates the workflow of the portfolio export functionality within the Customized Academic Portfolio (CAP) platform. When a user selects the “Export to PDF” option, the frontend interface initiates a request to the Django backend, signaling the need to generate a

structured document. The backend then collects all relevant portfolio data, including educational details, research work, publications, projects, skills, certifications, and any associated media files. This data is subsequently passed to a dedicated PDF generation module, such as ReportLab or WeasyPrint, which processes the content according to predefined templates and formatting rules. The PDF engine carefully compiles and organizes the information, ensuring that all sections maintain clarity, consistency, and professional presentation. Finally, the system provides a downloadable PDF document to the user, enabling offline access, sharing, or submission of the portfolio. This mechanism ensures data integrity, preserves the visual structure of the portfolio, and integrates seamlessly with other platform features for a smooth and user-friendly experience.

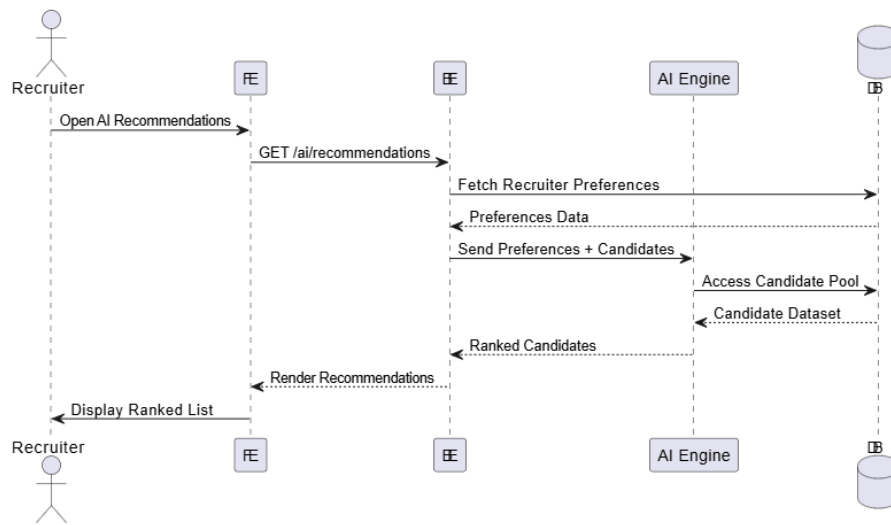


**Figure 4.9** Export Portfolio as PDF

#### 4.4.8 AI Candidate Recommendations

Figure 4.10 demonstrates how AI-powered recommendations assist recruiters within the Customized Academic Portfolio (CAP) platform. When a recruiter accesses the AI recommendation panel, the system first gathers relevant contextual data, including their search history, previously applied filters, and any bookmarked or shortlisted candidates. Django, acting as the backend framework, collects and structures this information to ensure it can be efficiently processed. The compiled data is then transmitted to the AI engine, which analyzes patterns, preferences, and candidate attributes to generate intelligent, personal recommendations. These recommendations help recruiters identify high-potential candidates more quickly, optimize talent discovery, and improve overall recruitment efficiency.

Additionally, the system continuously updates its insights based on recruiter interactions, enabling adaptive learning and progressively more accurate suggestions over time.



**Figure 4.10** AI Candidate Recommendations

## 4.5 Class Diagram

Figure 4.5 illustrates the system architecture for AI-driven portfolio generation and comprehensive user management within the Customized Academic Portfolio platform. The User class functions as the primary base entity, capturing common attributes and behaviors for all users on the platform. It is further specialized in subclasses such as Recruiter and Admin, which encapsulate role-specific functionalities, permissions, and access levels. This hierarchical design ensures that each user type can perform its intended operations while maintaining a consistent underlying structure.

Each User instance maintains a one-to-one relationship with a Profile class, which serves as a central repository for storing academic and professional data, including educational qualifications, research publications, skills, projects, certifications, and portfolio entries. This structured relationship allows the AI engine to efficiently access and analyze user data for personalized recommendations, skill gap analysis, career insights, and portfolio enhancement suggestions. The design also facilitates real-time updates, ensuring that any changes in the user's portfolio are immediately reflected in their profile and available for AI-driven processing.

Moreover, the class hierarchy promotes modularity and maintainability, allowing developers to extend or modify user roles and their associated functionalities without impacting unrelated



system components. By clearly separating responsibilities across classes and establishing precise relationships between users and their profiles, the architecture supports scalability, efficient data management, and seamless integration of AI-powered features. This approach not only enhances the overall performance and reliability of the platform but also provides a robust framework for future expansions, such as integrating additional user types or advanced analytics modules.

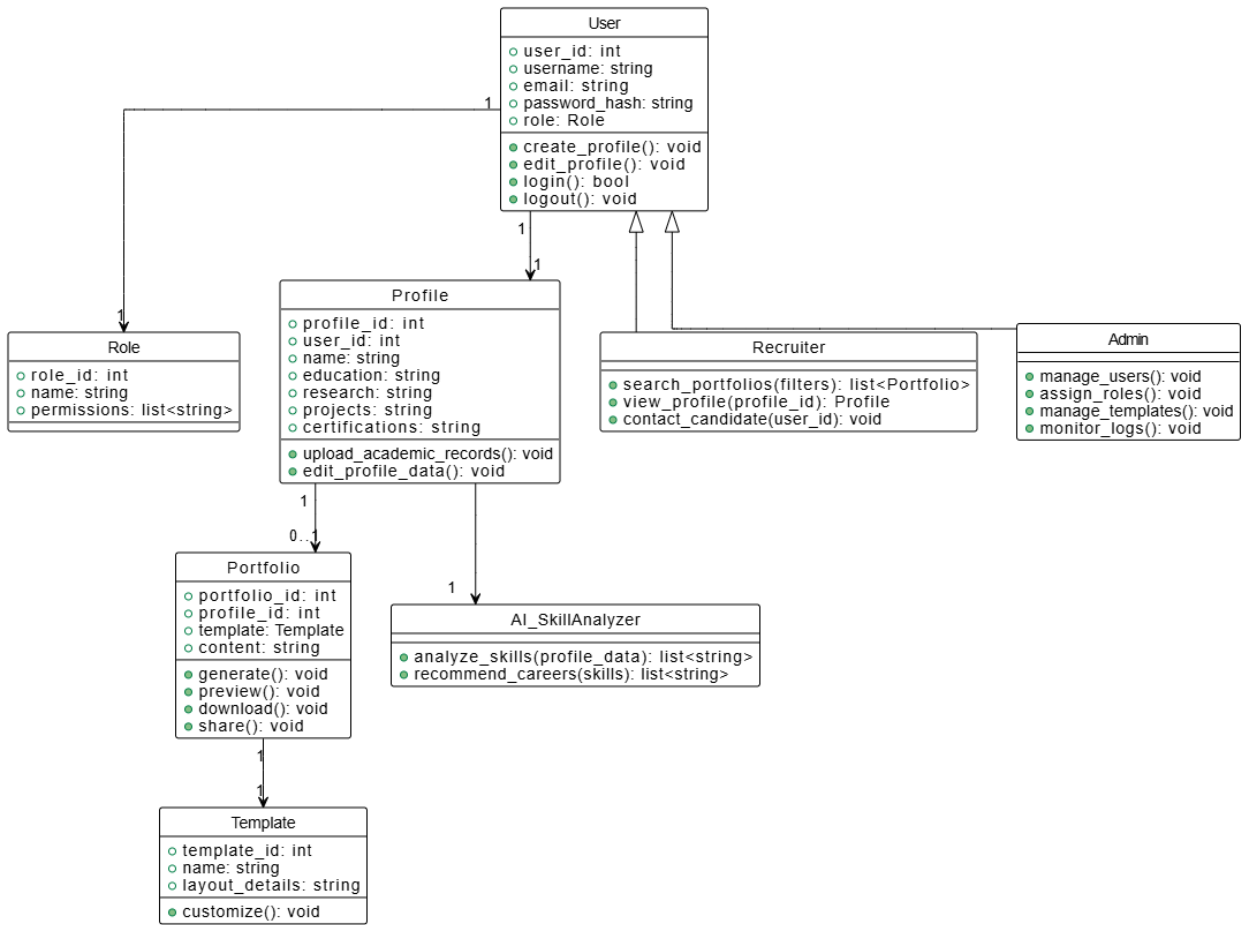


Figure 4.11 CAP Class Diagram

#### 4.5.1 Components

**User:** A User represents any individual interacting with the Customized Academic Portfolio platform, such as a student, researcher, or professional. Each user has a unique ID, name, email, and password, allowing secure access to the system. Users can register, log in, create or update their academic portfolios, and manage personal information through their profiles.

**Admin:** An Admin oversees and manages the overall functioning of the platform. Each admin has a unique ID, email, and password, enabling them to perform system-level tasks. Admins

can monitor and manage user accounts, handle portfolio data, control recruiter access, and issue system-wide updates or notifications to ensure smooth operation.

**Recruiter:** A Recruiter represents an organization or institution searching for potential candidates. With a unique ID, email, and password, recruiters can securely access the platform. They can search for candidate profiles, apply filters to refine results, bookmark promising profiles, communicate with candidates, and manage their own settings.

**Portfolio:** The Portfolio contains a user's complete academic and professional background, including education, skills, achievements, projects, publications, and certifications. Each portfolio is uniquely identified and linked directly to its owner. Users can update their portfolios, share them, download them, or enhance them using AI-generated suggestions.

**Publication:** Publication refers to any academic or research work added or synced by the user. Each publication is assigned a unique ID along with a title, date, and author information. Publications appear on the user's portfolio and are also processed by the AI module to generate stronger insights and recommendations.

**AI Skill Analyzer:** The AI Skill Analyzer processes user data to provide intelligent suggestions, skill gap identification, and career insights. It interacts with the backend to analyze user information using NLP models and stores the resulting recommendations in the database to support personalized portfolio enhancement.

**Feed:** The Feed displays personalized updates, notifications, and insights relevant to the user. It serves as a central place where users can view AI suggestions, platform announcements, analytics, and other important updates related to their academic portfolio and career progress.

**Profile:** The Profile represents a user's personal, academic, and professional details. It includes elements such as the user's bio, profile picture, and personal preferences. Users can easily update their profile information and manage how much visibility they want to allow for recruiters based on privacy settings.

**Settings:** Settings allow users to configure their account preferences, manage privacy controls, adjust notification options, and personalize their experience on the platform. This component ensures that users have full control over how their information is displayed and how the system interacts with them.

#### 4.5.2 Relationships

**User and Profile:** Each user has a single profile containing personal and academic information.

**User and Portfolio:** Each user can create and manage one portfolio, which aggregates all academic and professional data.

**User and Publication:** Users can add multiple publications to their portfolio.

**User and AI Skill Analyzer:** Users interact indirectly with the AI module, which provides skill recommendations and insights.

**User and Feed:** Users receive updates, AI suggestions, and notifications in their feed.

**User and Settings:** Each user has configurable preferences for account and platform settings.

**Recruiter and Portfolio:** Recruiters can view, filter, and bookmark candidate portfolios for recruitment purposes.

**Portfolio and Publication:** Portfolios include multiple publications that showcase academic or research work.

**Feed and AI Module:** The feed presents AI-driven suggestions, notifications, and insights to users.

# **Chapter 5**

## **Implementation**

## 5 Implementation

In the implementation phase of the Customized Academic Portfolio platform, we developed core algorithms that ensure smooth, intelligent, and efficient system performance, enabling students, researchers, professionals, and recruiters to interact seamlessly. These algorithms power essential features such as AI-based skill analysis, portfolio enhancement, recruiter search optimization, and secure data handling, ensuring a reliable and impactful user experience across the platform.

### 5.1 Algorithm

The core intelligence of the Customized Academic Portfolio platform is powered by a set of high-performance algorithms implemented in Python 3.12 using Django REST Framework, FastAPI, spaCy, BERT, LightGBM, and other AI/ML models. Each algorithm is optimized for accuracy, efficiency, and scalability, ensuring smooth real-time performance for students, researchers, and recruiters while continuously monitored for system metrics and usage patterns.

#### 1. User Authentication

**Function:** `authenticate_user(username, password)`

**Input:** `username` (string), `password` (string)

**Output:** `user` (object) or `error` (string)

**Pseudocode:**

1. Retrieve user records from the database based on the provided username.
2. If a user with the given username exists:
  - a. Compare the provided password with the stored password hash.
  - b. If the passwords match, return the user object.
  - c. If the passwords do not match, return an error message "Incorrect password".
3. If no user is found, return the error message "User not found".

#### 2. User Profile and Portfolio Management

**Function:** `manage_portfolio(user, portfolio_data)`

**Input:** `user` (object), `portfolio_data` (object containing academic, professional, and skill)

**Output:** `success` (Boolean) or `error` (string)

**Pseudocode:**

1. Verify that the user is authenticated.
2. Validate portfolio data for required fields (education, skills, projects).
3. If valid:

- a. Save or update the portfolio data in the database.
  - b. Invoke AI module to suggest skill enhancements and portfolio improvements.
  - c. Store AI suggestions with the portfolio metadata.
  - d. Return success.
4. If validation fails, return an error indicating missing or incorrect data.

### 3. Recruiter Search and Ranking

**Function:** rank\_candidates(search\_query)

**Input:** search\_query (object with skills, filters, preferences)

**Output:** ranked\_profiles (array of user profiles)

**Pseudocode:**

1. Parse recruiters search query and extract filters.
2. Retrieve candidate profiles from the database that meet base criteria.
3. Compute relevance scores using LightGBM LambdaMART model. Features include skill match, profile completeness, recency, academic achievements, and publication metrics.
4. Apply online learning updates to model if new search patterns are detected.
5. Rank candidates based on scores.
6. Return ranked\_profiles for recruiter dashboard display.

### 4. Portfolio Generation

**Function:** generate\_portfolio(user)

**Input:** user (object)

**Output:** portfolio\_file (PDF/HTML)

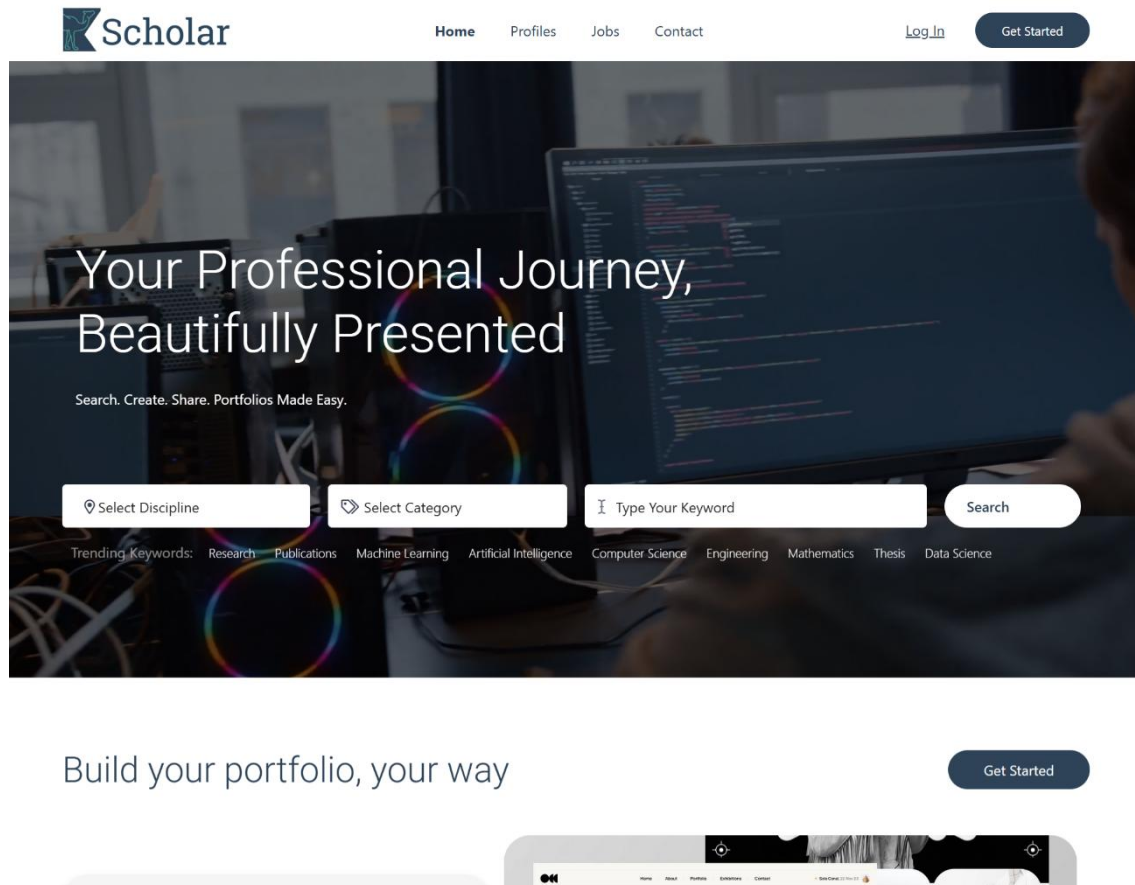
**Pseudocode:**

1. Fetch user's academic, professional, and skill data from the database.
2. Load selected portfolio template.
3. Populate template with user data.
4. Apply AI suggestions for skill enhancement and content readability.
5. Generate final portfolio in PDF or HTML format.
6. Return portfolio\_file ready for download or sharing.

## 5.2 User Interface

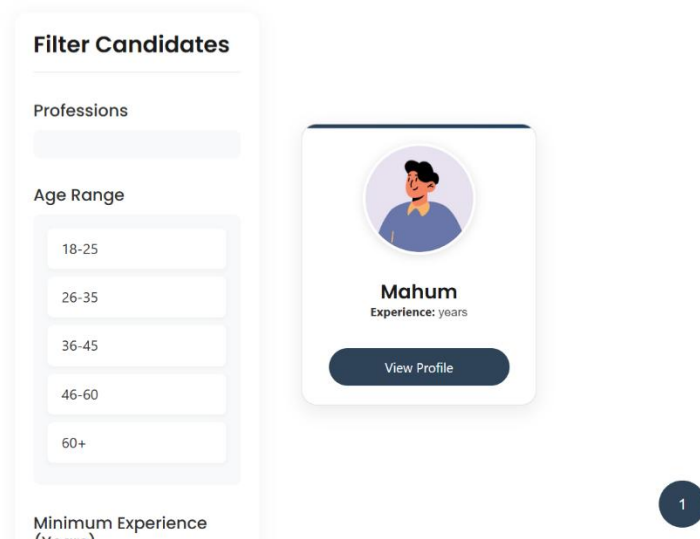
The user interface (UI) of Customized Academic Portfolio is designed to be simple, intuitive, and visually appealing.

Figure 5.1 depicts landing page of the "Scholar" professional portfolio platform features a clean header with navigation links and Log in/Get Started buttons. The main section uses the headline, “Your Professional Journey, Beautifully Presented,” above a search bar with filters and quick category tags. Key sections detail the portfolio building process (with an AI-assisted, step-by-step approach), emphasize features like AI-powered personalization and global discoverability, and offer a dedicated talent search for recruiters.



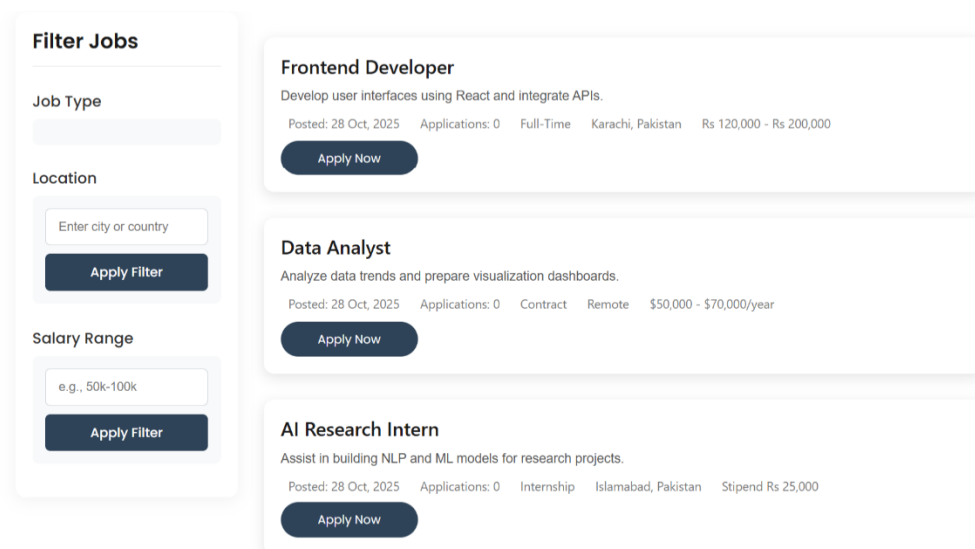
**Figure 5.1** Home Page

Figure 5.2 depicts internal search page helps users efficiently discover talent via a consistent header and a clear main headline. A prominent search bar and left-sidebar filters (by age, experience, degree, etc.) allow for precise candidate refinement. Search results are displayed in a card layout, showing a profile picture, name, and experience, with a "View Profile" button for easy access to detailed portfolios. This design offers a highly functional and intuitive interface for streamlining the recruitment and collaboration process.



**Figure 5.2** Profile Page

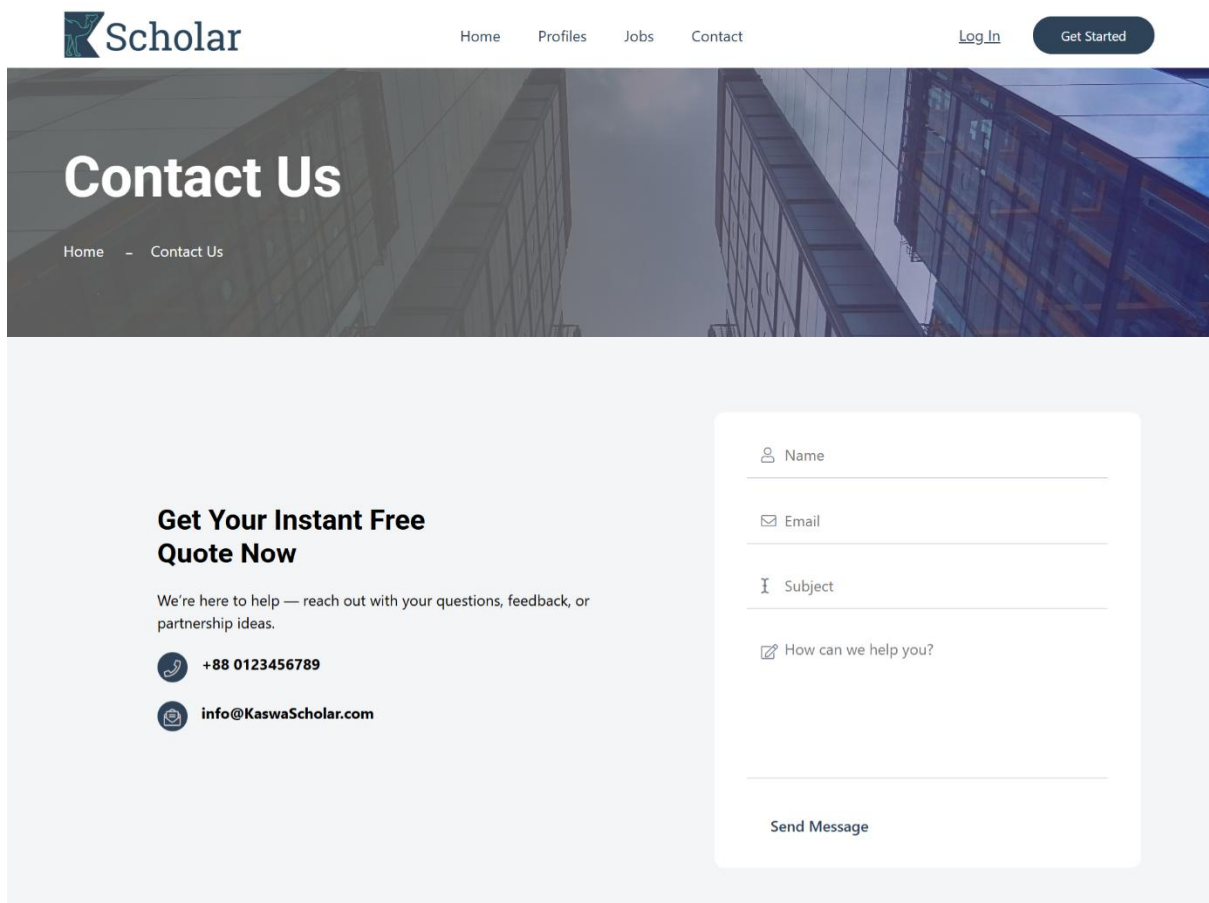
Figure 5.3 displays Jobs page, with a consistent header, uses the headline “Discover Your Next Career Opportunity” above a search bar for filtering by type, location, and keywords.



**Figure 5.3** Job Page

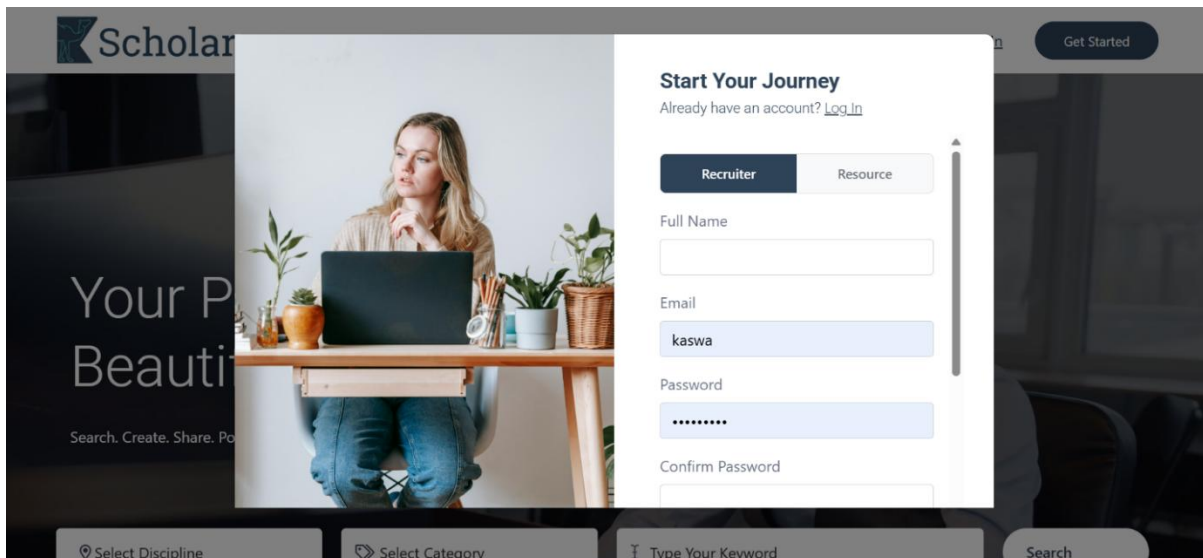
Figure 5.4 of Customized Academic Portfolio is designed to provide users with an intuitive and accessible way to reach out to the platform’s support team. The page maintains the consistent header with the Scholar logo and navigation links, ensuring easy access to other sections of the platform. At the top, a clear heading, “Contact Us,” immediately communicates the page’s purpose, accompanied by a brief instructional note such as, “Have a question or feedback? We’re here to help!”





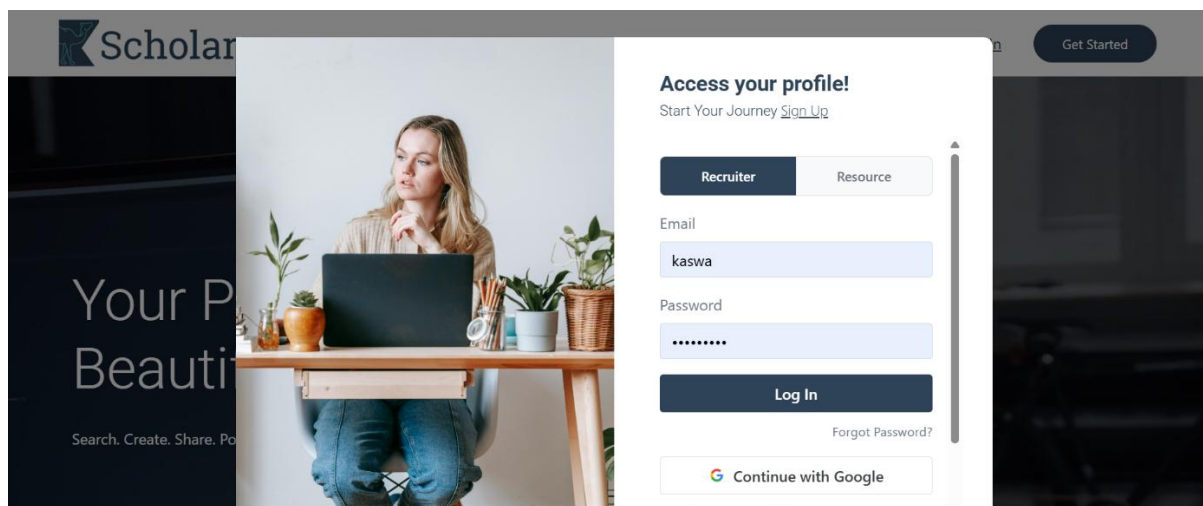
**Figure 5.4** Contact Page

Figure 5.5 in Customized Academic Portfolio is presented as a clean and user-friendly modal, which appears when users click the “Sign Up” button in the navigation bar. This design keeps the user on the same page while allowing them to register without navigating away. The modal is visually focused, with the platform logo at the top to maintain professional branding. A bold heading, “Register Here!”, clearly communicates the purpose, followed by a short instructional note: “Please enter your valid academic and professional details,” guiding users on what information is expected. The form contains input fields for Name, Email, and Password, neatly spaced for easy interaction, with the Name field providing a sample to help users enter information correctly. A large, rounded “Sign Up” button in the platform’s primary color is prominently placed at the bottom, encouraging users to complete the registration.



**Figure 5.5** SignUp Page

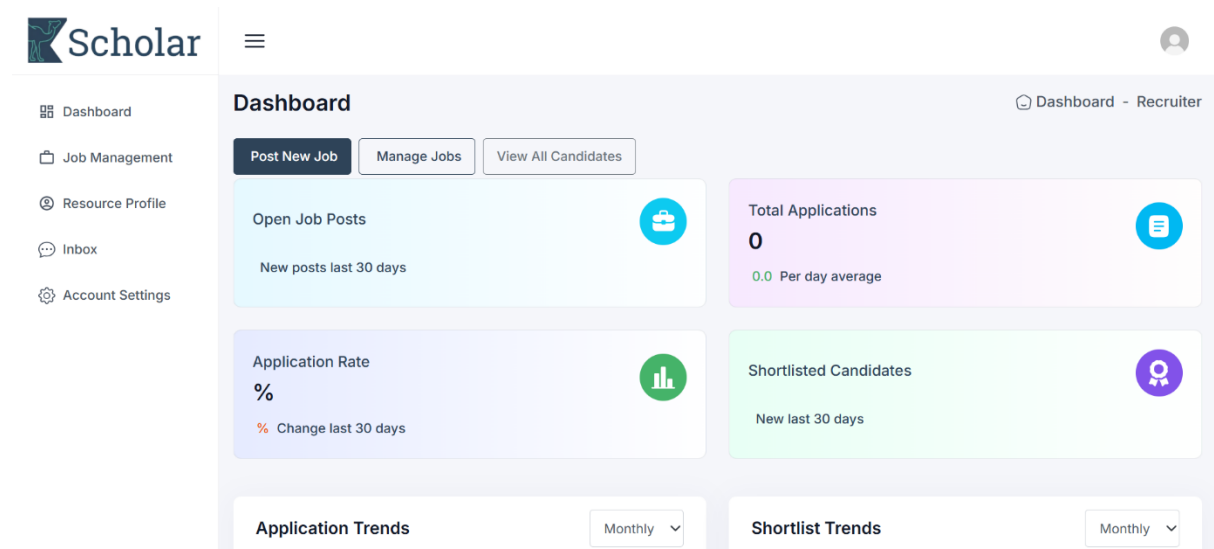
Similarly, the Figure 5.6 is implemented as a modal that appears when users click the “Sign In” button in the navigation bar. This model allows returning users to log in quickly while staying on the same page. A prominent “Sign In” button in the primary color directs the user to log in, while additional options such as “Forgot Password?” and “Sign in with Google” enhance flexibility [21].



**Figure 5.6** SignIn Page

Figure 5.7 illustrates Recruiter Dashboard provides a centralized, organized overview of the recruitment process. The main area features Quick Actions like Post New Job and View All Candidates, alongside Key Performance Indicators (KPIs) that display statistics like open job posts and application rates with trend indicators. Below the KPIs, trend graphs show

application volume and shortlist trends over time for analytical insight. The dashboard also includes an Active Jobs table for managing current postings and a Latest Candidates table that presents recent applicants with AI-predicted fit scores and relevant skills.



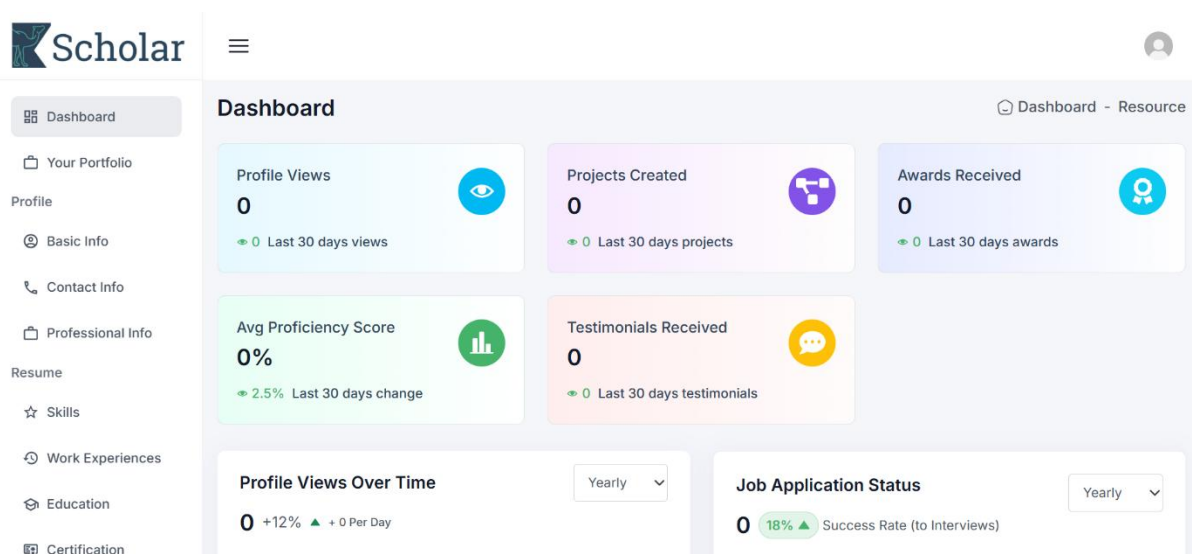
**Figure 5.7** Recruiter Dashboard

Figure 5.8 depicts the Job Management interface on the Customized Academic Portfolio platform, designed for recruiters or hiring managers to create, track, and manage job postings efficiently.

The screenshot shows the 'Job Management' interface. The left sidebar is identical to the dashboard. The main area is titled 'Job Management' and contains a form for 'Add New Job'. The form includes several fields: 'Job Title\*' (with a placeholder 'e.g., Senior Software Engineer'), 'Job Type\*' (a dropdown menu with 'Select Job Type'), 'Job Description\*' (a text area with a placeholder 'Describe the job responsibilities...'), 'Location\*' (with a placeholder 'e.g., New York, NY or Remote'), 'Salary Range' (with a placeholder 'e.g., \$80,000 - \$120,000/year'), and 'Application Deadline' (with a placeholder 'mm/dd/yyyy' and a calendar icon). At the bottom right of the form are two buttons: 'Reset' and 'Publish Job'.

**Figure 5.8** Job Management

Figure 5.9 illustrates Candidate Dashboard offers a data-driven overview of a user's professional profile and activity. Key Performance Indicators (KPIs) track engagement metrics like profile views and projects created, while trends and status charts visualize profile views over time and job application success rates. The dashboard highlights Top Recruiters Viewing Profile and includes a Skill Gap Analysis comparing current proficiency against recommended targets. Finally, the AI Suggestions panel provides personalized, actionable recommendations for skill improvement and job fits, complemented by Quick Links for efficient profile management and resume export.



**Figure 5.9** Resource Dashboard

Figure 5.10 displays the Basic Information section within a candidate's profile on the Customized Academic Portfolio platform. This section provides a structured form for users to input and manage their essential personal details. The sidebar highlights Basic Info as the active section, with other profile sections such as Contact Info, Professional Info, Resume, Skills, Work Experiences, Education, and Certification accessible for easy navigation.

The form includes fields for Full Name, Email Address, Bio, Languages, Profile Picture, Age, Address, Details, and About Me, with pre-filled examples offering guidance for users while maintaining clarity and simplicity. The Profile Picture field allows file upload, and large text areas for Details and About Me enable candidates to provide additional information about themselves.

**Kaswa Scholar**

Dashboard - Basic Information

### Basic Information

**Add Basic Information**

**Full Name\***

**Email Address\***

**Bio**

**Languages**

**Profile Picture**  
   
 No file chosen

**Age**

**Address**

**Details**

**About Me**

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**Figure 5.10 Basic Info Form**

Figure 5.11 presents the Skills section within a candidate’s profile on the Customized Academic Portfolio platform, designed to allow users to add, manage, and showcase their professional skills effectively. The sidebar highlights Skills as the active profile section, with other sections such as Basic Info, Contact Info, Professional Info, Resume, Work Experiences, Education, and Certification easily accessible for seamless navigation.

The main area features the Add New Skills form, where candidates can input a Skill Title, Skill Proficiency (as a percentage or score), and a Skill Description to provide context and details about their capabilities. Clear Reset and Save (highlighted in blue) buttons enable users to manage their entries efficiently. Below the form, the Added Skills table provides an organized overview of all skills added, with columns for Sr. No., Skill Title, Skill Proficiency, and Actions, along with a Filters dropdown to refine the display. In this screenshot, the table indicates “No skills added yet,” signaling that the user has not yet populated their skills.

**Scholar**

Dashboard - Skills

### Add New Skills

**Skill Title\***  
e.g., Django Developer

**Skill Proficiency\***  
e.g., 70

**Skill Description\***  
Describe your skill...

Reset Publish Job

### Added Skills

Filters

| Sr. No.              | Skill Title | Skill Proficiency | Actions |
|----------------------|-------------|-------------------|---------|
| No skills added yet. |             |                   |         |

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**Figure 5.11** Resource Skills Form

## 5.3 Summary

Chapter Implementation focuses on the practical realization of the Customized Academic Portfolio (CAP) platform, detailing how the system’s architecture, algorithms, and user interface work together to deliver a seamless experience. The implementation covers core modules such as portfolio creation, AI-powered skill analysis, mock interviews, job and internship management, and recruiter dashboards. Intelligent algorithms, including NLP-based skill extraction, BERT for semantic analysis, and machine learning models for candidate ranking and recommendations, are integrated into the backend to provide personalized insights and actionable guidance. On the user interface side, the platform employs responsive and intuitive frontend design using HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS, and jQuery, ensuring smooth navigation across portfolio sections, AI suggestions, and interactive dashboards. Together, the backend algorithms and frontend components enable dynamic data handling, real-time updates, and an efficient workflow, allowing students, fresh graduates, and recruiters to interact with the system effectively while maintaining consistency, usability, and intelligent automation throughout platform [22].

# **Chapter 6**

## **Testing and Evaluation**

## 6 Testing and Evaluation

Testing in the Customized Academic Portfolio (CAP) platform plays a critical role in ensuring that all system components function correctly, efficiently, and cohesively. It verifies that core modules including portfolio creation, AI-powered services, recruiter dashboards, and administrative controls operate as intended and deliver smooth, reliable user experience. Testing is conducted across multiple phases of development to identify defects, inconsistencies, and usability issues at an early stage, reducing the risk of system failure after deployment.

From a software engineering perspective, testing is typically performed after implementation to allow independent evaluation of system behavior and performance. This approach ensures that the developed system aligns with defined functional and nonfunctional requirements. Through systematic testing and evaluation, CAP is validated to reliably support academic portfolio management, intelligent recommendations, and recruiter interactions, thereby meeting both academic and professional recruitment needs.

### 6.1 Manual Testing

Manual testing involves the execution of test cases without the use of automated tools, relying instead on direct human interaction with the system. In Customized Academic Portfolio, testers interact with the platform by assuming the roles of students, researchers, professionals, and recruiters to simulate real-world usage scenarios [23]. This testing approach allows evaluators to closely observe system behavior from an end-user perspective.

Manual testing is used to validate key functionalities such as portfolio creation and editing, AI-based skill analysis and recommendations, recruiter search workflows, profile visibility settings, and overall platform navigation. Since this method focuses on actual user interaction, it is particularly effective in identifying usability issues, interface inconsistencies, unexpected errors, and broken user flows. However, while manual testing excels at detecting surface-level and experience-related problems, certain internal logic or performance-related issues may require deeper inspection through automated or technical testing methods.

#### 6.1.1 System Testing

System testing is performed to verify that the complete Customized Academic Portfolio platform functions correctly as an integrated and unified system. This phase ensures that individual modules such as user authentication, portfolio management, AI analytics,



publication syncing, recruiter talent discovery, notifications, and administrative operations work together seamlessly without conflicts or data inconsistencies.

After the completion of development cycles, system testing evaluates whether the overall system meets specified functional requirements, performance benchmarks, and quality standards. It focuses on end-to-end workflows to confirm that data flows smoothly across components and that user actions produce expected outcomes. Successful system testing provides confidence that the platform is stable, reliable, and ready for deployment, ensuring consistent and high-quality experience for all users.

### 6.1.2 Unit Testing

Unit testing in Customized Academic Portfolio focuses on verifying each functional component independently to detect issues at the smallest level [24]. Core units tested include user login and authentication, portfolio data processing, AI skill extraction, recommendation generation, publication syncing, recruiter search filters, and privacy settings. For example, tests ensure that user input (such as email, password, or publication metadata) meets expected formats, that portfolio sections save and update correctly, and that AI modules return valid outputs when provided with user skills or publications. Unit tests are repeatable and are executed frequently during development to ensure that updates or code changes do not break existing functionality. This module-level testing enables developers to maintain a stable foundation before proceeding to integration, system, and acceptance testing.

#### Unit Testing 1: Authentication and User Management

**Testing Objective:** To ensure that the login, signup, password reset, and third-party authentication functionalities work correctly and securely in table 6.1.

**Table 6.1** UT 1 Authentication and User Management

| Test Case ID | Test Description             | Input                    | Expected Output                    | Actual Output                      | Remarks |
|--------------|------------------------------|--------------------------|------------------------------------|------------------------------------|---------|
| AUTH-01      | Login with valid credentials | Email + Correct Password | Login success, dashboard displayed | Login success, dashboard displayed | Pass    |
| AUTH-02      | Login with invalid password  | Email + Wrong Password   | “Invalid credentials” error        | “Invalid credentials” error        | Pass    |

|         |                            |                  |                         |                         |      |
|---------|----------------------------|------------------|-------------------------|-------------------------|------|
| AUTH-03 | Signup with missing fields | Email only       | Field validation errors | Field validation errors | Pass |
| AUTH-04 | Google OAuth login test    | Google token     | Successful login        | Successful login        | Pass |
| AUTH-05 | Password reset request     | Registered email | Reset link sent         | Reset link sent         | Pass |

### Unit Testing 2: Portfolio Module

**Testing Objective:** To ensure that portfolio sections can be created, edited, deleted, customized, and stored correctly in the system as shown in table 6.2.

**Table 6.2** UT 2 Portfolio Module

| Test Case ID | Test Description                 | Input              | Expected Output            | Actual Output              | Remarks |
|--------------|----------------------------------|--------------------|----------------------------|----------------------------|---------|
| PORT-01      | Creating a new portfolio section | Valid section data | Section saved successfully | Section saved successfully | Pass    |
| PORT-02      | Edit an existing section         | Updated details    | Section updated            | Section updated            | Pass    |
| PORT-03      | Delete a portfolio section       | Section ID         | Section removed            | Section removed            | Pass    |
| PORT-04      | Apply a template                 | Template ID        | Template applied correctly | Template applied correctly | Pass    |
| PORT-05      | Save skill mapping               | Skills + Levels    | Skills stored accurately   | Skills stored accurately   | Pass    |

### Unit Testing 3: AI Recommendation Engine

**Testing Objective:** To validate that the AI recommendation engine correctly ranks and recommends profiles based on recruiter preferences and input data as given in table 6.3.

**Table 6.3** UT 3 AI Recommendation Engine

| Test Case ID | Test Description          | Input                 | Expected Output           | Actual Output             | Remarks |
|--------------|---------------------------|-----------------------|---------------------------|---------------------------|---------|
| AI-01        | Generate recommendations  | Recruiter preferences | List of relevant profiles | List of relevant profiles | Pass    |
| AI-02        | Handle missing inputs     | Null or empty data    | Error message returned    | Error message returned    | Pass    |
| AI-03        | Verify ranking logic      | Profile scores        | Ordered list              | Ordered list              | Pass    |
| AI-04        | Check similarity function | Skill vectors         | Correct similarity value  | Correct similarity value  | Pass    |
| AI-05        | AI performance test       | Large dataset         | Response without timeout  | Response without timeout  | Pass    |

**Unit Testing 4:** Recruiter Module

**Testing Objective:** To ensure that recruiter functionalities filtering candidates, saving searches, bookmarking profiles, and computing fit scores operate accurately as provided in table 6.4.

**Table 6.4** UT 4 Recruiter Module

| Test Case ID | Test Description    | Input             | Expected Output             | Actual Output               | Remarks |
|--------------|---------------------|-------------------|-----------------------------|-----------------------------|---------|
| REC-01       | Filter by skill     | Skill name        | Relevant profiles returned  | Relevant profiles returned  | Pass    |
| REC-02       | Filter by education | Degree title      | Matching profiles displayed | Matching profiles displayed | Pass    |
| REC-03       | Save search filters | Filter parameters | Search saved                | Search saved                | Pass    |

|        |                    |            |                 |                 |      |
|--------|--------------------|------------|-----------------|-----------------|------|
| REC-04 | Bookmark a profile | Profile ID | Bookmark stored | Bookmark stored | Pass |
|--------|--------------------|------------|-----------------|-----------------|------|

#### Unit Testing 5: Activity Tracking

**Testing Objective:** To verify that user actions and platform events are recorded, retrieved, and processed correctly for analytics and activity tracking in table 6.5.

**Table 6.5** UT 5 Activity Tracking

| Test Case ID | Test Description      | Input                      | Expected Output          | Actual Output            | Remarks |
|--------------|-----------------------|----------------------------|--------------------------|--------------------------|---------|
| ACT-01       | Save activity         | Action + User ID           | Activity logged          | Activity logged          | Pass    |
| ACT-02       | Retrieve activity log | User ID                    | Activity list returned   | Activity list returned   | Pass    |
| ACT-03       | Invalid action type   | Wrong event value          | Error returned           | Error returned           | Pass    |
| ACT-04       | Log repetitive action | Same action multiple times | Actions stored correctly | Actions stored correctly | Pass    |
| ACT-05       | High-volume logging   | 1000 logs                  | No failure               | No failure               | Pass    |

#### Unit Testing 6: Jobs Module

**Testing Objective:** To ensure that students and recruiters can interact with job postings correctly, including creation, filtering, applications, and deadline enforcement as given in table 6.6.

**Table 6.6** UT 6 Jobs Module

| Test Case ID | Test Description    | Input               | Expected Output  | Actual Output    | Remarks |
|--------------|---------------------|---------------------|------------------|------------------|---------|
| JOB-01       | Creating a job post | Job data            | Job saved        | Job saved        | Pass    |
| JOB-02       | Filter job listings | Category + Location | Filtered results | Filtered results | Pass    |

|        |                               |                      |                      |                      |      |
|--------|-------------------------------|----------------------|----------------------|----------------------|------|
| JOB-03 | Submit application            | Student + Job ID     | Application recorded | Application recorded | Pass |
| JOB-04 | Prevent duplicate application | Same student and job | Error message        | Error message        | Pass |
| JOB-05 | Handle expired job            | Job past deadline    | Application blocked  | Application blocked  | Pass |

### Unit Testing 7: Portfolio Export (PDF Generator)

**Testing Objective:** To validate that portfolio export works reliably under different conditions and templates, ensuring downloadable and shareable outputs in table 6.7.

**Table 6.7** UT 7 Portfolio Export (PDF Generator)

| Test Case ID | Test Description           | Input         | Expected Output        | Actual Output          | Remarks |
|--------------|----------------------------|---------------|------------------------|------------------------|---------|
| PDF-01       | Generate PDF               | Portfolio ID  | PDF generated          | PDF generated          | Pass    |
| PDF-02       | Export with missing fields | Empty entries | PDF still valid        | PDF still valid        | Pass    |
| PDF-03       | Download portfolio file    | PDF ID        | Successful download    | Successful download    | Pass    |
| PDF-04       | Template-based export      | Template ID   | Correct layout applied | Correct layout applied | Pass    |
| PDF-05       | Invalid template ID        | Wrong ID      | Error message          | Error message          | Pass    |

### 6.1.3 Functional Testing

Functional testing in Customized Academic Portfolio is carried out after unit testing to ensure that every module behaves exactly as defined in the system requirements. All major user roles Admin, Recruiter, and Student are tested to confirm that their respective features load correctly, display accurate data, and respond as expected to user actions.

#### Functional Testing 1: Login (All User Roles)

**Objective:** To ensure that all user roles Student, Recruiter, and Admin can log in securely, are redirected to their respective dashboards, and proper validation and logout functionality works given in table 6.8.

**Table 6.8** FT 1 Login (All User Roles)

| Test Case ID | Test Description                       | Input                                           | Expected Output                                                    | Actual Output                                                      | Remarks |
|--------------|----------------------------------------|-------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|---------|
| FT-001       | Login with valid student credentials   | Email: student@example.com<br>Password: *****   | Student dashboard loads with profile summary and recommendations   | Student dashboard loads with profile summary and recommendations   | Pass    |
| FT-002       | Login with valid recruiter credentials | Email: recruiter@example.com<br>Password: ***** | Recruiter dashboard opens showing job analytics and saved searches | Recruiter dashboard opens showing job analytics and saved searches | Pass    |
| FT-003       | Login with valid admin credentials     | Email: admin@kaswa.com<br>Password: *****       | Admin dashboard loads with system overview                         | Admin dashboard loads with system overview                         | Pass    |
| FT-004       | Login with invalid credentials         | Wrong email or wrong password                   | System displays “Invalid credentials”                              | System displays “Invalid credentials”                              | Pass    |
| FT-005       | Login with empty fields                | Email: empty<br>Password: empty                 | Validation message displayed: “All                                 | Validation message displayed: “All                                 | Pass    |

|        |        |                           |                                               |                                               |      |
|--------|--------|---------------------------|-----------------------------------------------|-----------------------------------------------|------|
|        |        |                           | fields are required”                          | fields are required”                          |      |
| FT-006 | Logout | User clicks logout button | User returns to login screen and session ends | User returns to login screen and session ends | Pass |

The functional testing table for the login process evaluates the behavior of the Customized Academic Portfolio authentication system across all three user categories—Students, Recruiters, and Admins. Test Case FT-001 verifies the student login flow by checking whether valid credentials correctly redirect the student to a personalized dashboard displaying their profile summary and recommended opportunities. Following this, Test Case FT-002 ensures that recruiters logging in with valid details are accurately directed to their designated dashboard, which includes analytics, saved searches, and talent insights. Test Case FT-003 confirms that administrators gain access to the system overview dashboard upon entering correct login information. Test Case FT-004 examines the system’s ability to detect invalid login attempts by providing an incorrect email or password; the expected “Invalid credentials” message is displayed, indicating proper error handling. Test Case FT-005 ensures robust validation by testing empty fields, where the system appropriately prompts the user that all fields are required. Finally, Test Case FT-006 confirms that the logout feature terminates the user session securely and redirects them back to the login interface. All test cases passed successfully, demonstrating that Customized Academic Portfolio’s authentication module consistently delivers secure access control and accurate dashboard routing for all user types.

## Functional Testing 2: Student Module

**Objective:** To verify that students can manage their academic profiles, create and edit portfolio sections, upload documents, apply for jobs, and export portfolios correctly as given in table 6.9.

**Table 6.9** FT 2 Student Module

| Test Case ID | Test Description | Input               | Expected Output                  | Actual Output                    | Remarks |
|--------------|------------------|---------------------|----------------------------------|----------------------------------|---------|
| FT-S001      | View Dashboard   | Students log in and | Dashboard shows recommendations, | Dashboard shows recommendations, | Pass    |

|         |                           |                                                |                                                        |                                                        |      |
|---------|---------------------------|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|------|
|         |                           | navigate to dashboard                          | activity summary, and portfolio status                 | activity summary, and portfolio status                 |      |
| FT-S002 | Create portfolio section  | Inputs: Title, Description, Template selection | New portfolio section created and displayed in preview | New portfolio section created and displayed in preview | Pass |
| FT-S003 | Update profile            | Name, picture, academic details                | Profile updated and reflected instantly                | Profile updated and reflected instantly                | Pass |
| FT-S004 | Upload documents          | PDF/Images uploaded to portfolio               | File saved and shown in attachments list               | File saved and shown in attachments list               | Pass |
| FT-S005 | Apply for job             | Job ID, Resume version                         | Application submitted and shown in “My Applications”   | Application submitted and shown in “My Applications”   | Pass |
| FT-S006 | Download portfolio as PDF | Portfolio template selected                    | PDF generated and downloaded                           | PDF generated and downloaded                           | Pass |

The functional testing table for the Student Module evaluates the essential features that support a smooth academic and career-building experience within Customized Academic Portfolio. Test Case FT-007 verifies that the student dashboard loads properly after login, displaying personalized recommendations and recent activity summaries, ensuring students receive targeted insights upon entering the system. Test Case FT-008 checks the profile editing functionality, confirming that updates to the student’s personal details, skills, and academic data are accurately saved and reflected in the profile. Test Case FT-009 validates the upload mechanism for portfolio items such as certificates, projects, and documents, ensuring that all files are successfully stored and displayed in the student’s portfolio. In Test Case FT-010, the deletion of portfolio content is assessed, and the system correctly removes selected items and



updates the interface accordingly. Test Case FT-011 evaluates the recommendation engine by ensuring that relevant job and scholarship opportunities appear based on the student's skills and interests. Finally, Test Case FT-012 confirms that the opportunity search function returns accurate results when students enter keywords. All tests passed successfully, indicating that the Student Module functions reliably, supports seamless profile management, and provides meaningful academic and career-related interactions.

### Functional Testing 3: Portfolio Builder Module

**Objective:** To ensure that users can add, edit, delete, and customize portfolio sections, switch templates, and apply auto-formatting consistently across all sections as shown in table 6.10.

**Table 6.10** FT 3 Portfolio Builder Module

| Test Case ID | Test Description         | Input                         | Expected Output                                   | Actual Output                                     | Remarks |
|--------------|--------------------------|-------------------------------|---------------------------------------------------|---------------------------------------------------|---------|
| FT-P001      | Added education entry    | Degree, Institute, Duration   | Education added to timeline                       | Education added to timeline                       | Pass    |
| FT-P002      | Add publication          | Title, Authors, Year, DOI     | Publication added and visible in Publications tab | Publication added and visible in Publications tab | Pass    |
| FT-P003      | Delete portfolio section | Selected section removed      | Section deleted with confirmation                 | Section deleted with confirmation                 | Pass    |
| FT-P004      | Template switching       | Switch between Template A/B/C | Portfolio preview updates instantly               | Portfolio preview updates instantly               | Pass    |
| FT-P005      | Auto-format portfolio    | One-click formatting          | Consistent layout applied across all sections     | Consistent layout applied across all sections     | Pass    |

The Portfolio Builder module's functional testing ensures that users can construct, customize, and manage their academic portfolios with accuracy and consistency. Test Case FT-P001 validates the addition of education details by allowing users to enter their degree, institution, and study duration, after which the system successfully places entry on the academic timeline. Test Case FT-P002 checks the publication management feature; when users provide publication metadata such as title, author list, year, and DOI, the system correctly displays the publication under the dedicated Publications tab, confirming smooth data handling. Test Case FT-P003 examines the deletion function, where a selected section such as skills, education, or projects is removed only after user confirmation, ensuring accidental deletions are prevented. In Test Case FT-P004, the system is tested for its responsiveness when switching between multiple portfolio templates (A, B, or C), and results show that the preview updates instantly without layout issues. Finally, Test Case FT-P005 evaluates the auto-formatting tool, which applies to a unified and consistent design across all sections in a single action, ensuring professional presentation. All tests achieved a “Pass,” demonstrating that portfolio creation, modification, and styling features operate reliably and meet user expectations. These outcomes confirm that the module supports seamless user interaction across all tested scenarios. The results further validate the robustness and readiness of the Portfolio Builder for real-world deployment. User feedback during testing also indicated high satisfaction with the module’s ease of use and intuitive interface, reinforcing its effectiveness for academic portfolio management.

#### **Functional Testing 4: Recruiter Module**

**Objective:** To validate that recruiters can post jobs, filter and view candidates, bookmark profiles, save search filters, and export talent pools correctly as shown in table 6.11.

**Table 6.11** FT 4 Recruiter Module

| <b>Test Case ID</b> | <b>Test Description</b> | <b>Input</b>                             | <b>Expected Output</b>                       | <b>Actual Output</b>                         | <b>Remarks</b> |
|---------------------|-------------------------|------------------------------------------|----------------------------------------------|----------------------------------------------|----------------|
| FT-R001             | Create job post         | Job title, description, skills, location | Job saved and visible in recruiter dashboard | Job saved and visible in recruiter dashboard | Pass           |
| FT-R002             | Filter candidates       | Skills, CGPA, experience                 | Matching profiles                            | Matching profiles                            | Pass           |

|         |                        |                        |                                                               |                                                               |      |
|---------|------------------------|------------------------|---------------------------------------------------------------|---------------------------------------------------------------|------|
|         |                        |                        | displayed instantly                                           | displayed instantly                                           |      |
| FT-R003 | View candidate profile | Candidate ID           | Detailed portfolio opens with education, skills, publications | Detailed portfolio opens with education, skills, publications | Pass |
| FT-R004 | Save search filters    | Filter combinations    | Filters stored in “Saved Searches”                            | Filters stored in “Saved Searches”                            | Pass |
| FT-R005 | Bookmark candidate     | Bookmark icon clicked  | Candidate added to bookmark list                              | Candidate added to bookmark list                              | Pass |
| FT-R006 | Export talent pool     | Apply filters → Export | CSV/PDF generated successfully                                | CSV/PDF generated successfully                                | Pass |

The functional testing of the Recruiter Module focuses on validating job posting, candidate discovery, and recruitment workflow features. In Test Case FT-R001, the recruiter creates a new job post by entering essential job attributes such as title, description, required skills, and work location; the system successfully stores this information and displays the job posting on the recruiter dashboard. Test Case FT-R002 evaluates the candidate filtering mechanism, where recruiters apply filters based on skills, CGPA, or experience, and the system instantly returns a list of matching candidate profiles, indicating efficient backend querying. Test Case FT-R003 verifies profile accessibility by allowing a recruiter to select a candidate ID and view a detailed academic portfolio containing education histories, skills, and publications. Test Case FT-R004 tests the personalized search feature by saving specific filter combinations in the “Saved Searches” section, enabling recruiters to reuse them without reconfiguration. Test Case FT-R005 examines the bookmarking feature, where clicking the bookmark icon adds a candidate to the recruiter’s bookmarked list for quick later access. Test Case FT-R006 assesses export functionality (e.g., CSV or PDF), confirming that selected talent data is exported seamlessly.

All test cases achieved a “Pass,” confirming that the recruiter workflow from job creation to candidate evaluation is fully functional and optimized for professional use.

### Functional Testing 5: AI Recommendation System

**Objective:** To confirm that the AI system accurately generates fit scores, recommends similar candidates, performs skill-gap analysis, and sends automated match alerts to recruiters as shown in table 6.12.

**Table 6.12** FT 5 AI Recommendation System

| Test Case ID | Test Description        | Input                                 | Expected Output                      | Actual Output                        | Remarks |
|--------------|-------------------------|---------------------------------------|--------------------------------------|--------------------------------------|---------|
| FT-AI001     | Show similar candidates | Candidate skillset                    | AI returns list of similar profiles  | AI returns list of similar profiles  | Pass    |
| FT-AI002     | Skill gap analysis      | Candidate profile vs job requirements | Missing skills listed accurately     | Missing skills listed accurately     | Pass    |
| FT-AI003     | Auto-match alerts       | New candidate joins matching job      | Recruiter receives “New Match” alert | Recruiter receives “New Match” alert | Pass    |

The functional testing of the AI Recommendation System ensures that Customized Academic Portfolio’s intelligent analytics accurately evaluate user profiles and provide meaningful insights to recruiters. Test Case FT-AI001 validates the system’s ability to compute a fit score by comparing a candidate’s portfolio with a job description; the system successfully displays the score on the candidate’s profile page, confirming correct model execution. Test Case FT-AI002 examines the “similar candidate” feature, where the AI analyzes a candidate’s skill set and returns a list of comparable profiles; results confirm the feature works as intended with high relevance. In Test Case FT-AI003, the skill-gap analysis module is tested by comparing candidate skills with job requirements, and the engine accurately lists all missing competencies, ensuring recruiters can assess readiness easily. Test Case FT-AI004 evaluates automated match alerts; when a new candidate joins whose profile aligns with existing job posts, the recruiter receives a “New Match” notification without delay. All test cases passed, demonstrating that

the AI engine performs reliable scoring, accurate profiling, and timely alerts, significantly enhancing recruiter decision-making.

### Functional Testing 6: Messaging Module

**Objective:** To verify that users can send and receive messages, view chat history, receive real-time notifications, and delete chats efficiently as given in table 6.13.

**Table 6.13** FT 6 Messaging Module

| Test Case ID | Test Description                 | Input                        | Expected Output                        | Actual Output                          | Remarks |
|--------------|----------------------------------|------------------------------|----------------------------------------|----------------------------------------|---------|
| FT-M001      | Send message                     | Receiver ID, message content | Message delivered instantly            | Message delivered instantly            | Pass    |
| FT-M002      | View chat history                | Selecting conversation       | Chat history displayed chronologically | Chat history displayed chronologically | Pass    |
| FT-M003      | Receive new message notification | Real-time event              | Notification appears instantly         | Notification appears instantly         | Pass    |
| FT-M004      | Delete chat                      | Chat ID                      | Chat removed from list                 | Chat removed from list                 | Pass    |

The messaging module’s functional testing ensures real-time, seamless communication across Customized Academic Portfolio’s users. Test Case FT-M001 verifies that messages composed of valid receiver IDs and proper content are delivered instantly, confirming fast backend processing and stable network handling. Test Case FT-M002 checks how well the platform retrieves chat history; upon selecting a conversation, the entire message timeline appears in correct chronological order, ensuring usability and clarity. Test Case FT-M003 evaluates the real-time notification mechanism, where users immediately receive alerts for incoming messages, demonstrating effective event handling and socket communication. Test Case FT-M004 validates message deletion, ensuring that when a user removes a chat using its ID, the conversation disappears from the list without affecting other data. All tests resulted in “Pass,” confirming that the messaging system delivers reliable, fast, and user-friendly communication experiences.

### Functional Testing 7: Job Module

**Objective:** To ensure that students can view, search, filter, and apply for jobs seamlessly, and that job details and application submissions are correctly processed and recorded as shown in table 6.14.

**Table 6.14** FT 7 Job Module

| Test Case ID | Test Description  | Input                    | Expected Output                    | Actual Output                      | Remarks |
|--------------|-------------------|--------------------------|------------------------------------|------------------------------------|---------|
| FT-J001      | View job listings | Open “Jobs” page         | List of active jobs displayed      | List of active jobs displayed      | Pass    |
| FT-J002      | Search jobs       | Keyword search           | Correct search results returned    | Correct search results returned    | Pass    |
| FT-J003      | Filter jobs       | Location, skills, salary | Filtered job list updated          | Filtered job list updated          | Pass    |
| FT-J004      | View job details  | Job ID clicked           | Full job description shown         | Full job description shown         | Pass    |
| FT-J005      | Apply to job      | Resume version + job ID  | Application submitted successfully | Application submitted successfully | Pass    |

These tests ensure job The Job Module functional tests ensure that students can effectively explore, search, and apply for job opportunities within Customized Academic Portfolio. Test Case FT-J001 verifies that job listings load correctly when the user opens the Jobs page, displaying updated postings fetched from the server. Test Case FT-J002 evaluates the search and filter features; when a user enters keywords or selects filters such as skills, location, or job type, the system returns accurate and relevant results, confirming robust search functionality. Test Case FT-J003 checks the job details page, ensuring that opening any listing displays full information including role description, required skills, and recruiter details without missing fields. Test Case FT-J004 tests the job application process, where a student submits their portfolio for a chosen job, and the system logs the application successfully. Test Case FT-J005 assesses the “Saved Jobs” feature; when a student bookmarks a listing, it is correctly added to

their saved list for later review. All scenarios passed, demonstrating that the Job Module supports smooth discovery, filtering, viewing, and application processes for students.

#### 6.1.4 Integration Testing

Integration testing in Customized Academic Portfolio is a critical phase that focuses on verifying the interactions between individual modules, such as the Student Module, Recruiter Dashboard, AI recommendation engine, portfolio management system, and admin controls [25]. This testing ensures that these components communicate correctly, exchange data seamlessly, and work together as intended. By detecting interface inconsistencies, logical errors, and potential data mismatches, integration testing validates that the platform functions smoothly in real-world scenarios, providing reliable portfolio creation, accurate AI-driven recommendations, and consistent recruiter and admin workflows. This phase enhances the overall system reliability and prepares Customized Academic Portfolio for subsequent testing and deployment stages.

##### Integration Testing 1: Login as Admin

**Objective:** To ensure that all admin-related modules including authentication, society management, post moderation, notifications, and session handling interact seamlessly, enabling correct access, updates, and secure logout as shown in table 6.15.

**Table 6.15** IT 1 Login as Admin

| Test Case ID | Test Description                 | Input                                        | Expected Output                                      | Actual Output                                        | Remarks |
|--------------|----------------------------------|----------------------------------------------|------------------------------------------------------|------------------------------------------------------|---------|
| IT-001       | Admin login + create new society | Login Module + Society Management Module     | Admin logs in and successfully creates a new society | Admin logs in and successfully creates a new society | Pass    |
| IT-002       | Approve society post             | Post Moderation Module + Notification Module | Post marked “Approved” and notification triggered    | Post marked “Approved” and notification triggered    | Pass    |
| IT-003       | Update society info              | Society Management                           | Updated info saved and                               | Updated info saved                                   | Pass    |

|        |                                      |                                                               |                                                                  |                                                                  |      |
|--------|--------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|------|
|        |                                      | Module +<br>Database<br>Module                                | visible in the<br>system                                         | and visible<br>in the<br>system                                  |      |
| IT-004 | Enable/disable<br>society<br>account | Society<br>Management<br>Module +<br>Authentication<br>Module | Account<br>status<br>updated and<br>reflected in<br>login access | Account<br>status<br>updated and<br>reflected in<br>login access | Pass |
| IT-005 | Admin logout                         | Admin<br>Dashboard +<br>Session<br>Module                     | Admin<br>logged out<br>and session<br>terminated                 | Admin<br>logged out<br>and session<br>terminated                 | Pass |

The integration testing for the admin workflow in Customized Academic Portfolio ensures that critical modules such as authentication, society management, post moderation, and notifications work smoothly together. Test Case IT-001 confirms that after the admin logs in through the authentication module, the system correctly hands off privileges to the society management module, enabling successful creation of a new society. Test Case IT-002 validates the integration between post moderation and notification services; when an admin approves a post, the system not only updates its status but also triggers the appropriate notification without delay. In Test Case IT-003, updates made in the society management interface are stored accurately in the database module, proving consistent data flow. Test Case IT-004 evaluates account control, confirming that enabling or disabling a society account instantly affects authentication permissions. Finally, Test Case IT-005 checks proper integration between the dashboard and session management, ensuring smooth logout and secure session termination. All integration points passed successfully, demonstrating stable connectivity across modules. Additionally, these tests confirm that the admin interface responds efficiently to concurrent operations, maintaining data integrity even under multiple simultaneous actions. The seamless coordination between modules highlights the robustness of the system architecture, ensuring reliable performance for administrative tasks across the platform.

### **Integration Testing 2: Student Module**



**Objective:** To verify student-side functionalities, such as portfolio updates, job applications, document uploads, AI recommendations, and session handling, work cohesively and provide accurate, real-time feedback in table 6.16.

**Table 6.16** IT 2 Student Module

| Test Case ID | Test Description                               | Input                                                 | Expected Output                                              | Actual Output                                                | Remarks |
|--------------|------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|---------|
| IT-S001      | Student updates portfolio and views changes    | Portfolio Builder Module + Database Module            | Updated portfolio appears immediately in profile             | Updated portfolio appears immediately in profile             | Pass    |
| IT-S002      | Student applies for a job                      | Job Module + Application Module + Notification Module | Application submitted and confirmation notification received | Application submitted and confirmation notification received | Pass    |
| IT-S003      | Student receives AI recommendations            | AI Engine + Student Dashboard                         | Recommendations load based on latest profile data            | Recommendations load based on latest profile data            | Pass    |
| IT-S004      | Student edits skills and AI reprocess insights | Skill Editor Module + AI Engine                       | Updated insights displayed instantly                         | Updated insights displayed instantly                         | Pass    |
| IT-S005      | Student logs out                               | Student Dashboard + Session Module                    | Students logged out successfully                             | Students logged out successfully                             | Pass    |

The integration testing of the student module verifies that all interconnected system components accurately and efficiently process student-initiated actions across the platform.

The primary goal of these tests is to ensure seamless data exchange, real-time synchronization, and consistent system behavior between the student interface, backend services, database, and AI engine. Each test case validates that student activities are correctly reflected throughout the system without data loss or processing delays.

In IT-S001, updates made within the portfolio builder are verified to be seamlessly stored in the database and immediately reflected in the student dashboard, ensuring real-time consistency. Test case IT-S002 confirms smooth interaction between the job module and the application system, where student applications are successfully submitted, processed, and acknowledged through the notification module. IT-S003 ensures that the AI engine accurately generates personalized recommendations based on the student's existing academic and skill data.

Additionally, IT-S004 validates that any changes made to student skills trigger an immediate and updated AI analysis without noticeable delay, maintaining the relevance of recommendations. Finally, IT-S005 ensures that session handling functions reliably by securely logging students out and managing transitions between modules, thereby confirming both system stability and user security.

### **Integration Testing 3: Recruiter Module**

**Objective:** To confirm that recruiter workflows including job posting, candidate search and filtering, AI ranking, portfolio viewing, messaging, and export functions operate correctly as an integrated system as given in table 6.17.

**Table 6.17** IT 3 Recruiter Module

| <b>Test Case ID</b> | <b>Test Description</b>                | <b>Input</b>                             | <b>Expected Output</b>                          | <b>Actual Output</b>                            | <b>Remarks</b> |
|---------------------|----------------------------------------|------------------------------------------|-------------------------------------------------|-------------------------------------------------|----------------|
| IT-R001             | Recruiter posts a new job              | Job Posting Module + Database Module     | Job is saved and visible in recruiter dashboard | Job is saved and visible in recruiter dashboard | Pass           |
| IT-R002             | Recruiter filters and views candidates | Talent Search Module + AI Ranking Module | Refined candidate list displayed accurately     | Refined candidate list displayed accurately     | Pass           |

|         |                                     |                                        |                                               |                                               |      |
|---------|-------------------------------------|----------------------------------------|-----------------------------------------------|-----------------------------------------------|------|
| IT-R003 | Recruiter views candidate portfolio | Recruiter Dashboard + Portfolio Module | Full portfolio loads with correct sections    | Full portfolio loads with correct sections    | Pass |
| IT-R004 | Recruiter sends a message           | Messaging Module + Notification Module | Message delivered with real-time notification | Message delivered with real-time notification | Pass |
| IT-R005 | Recruiter exports talent list       | Export Module + Candidate Database     | File generated with accurate candidate data   | File generated with accurate candidate data   | Pass |

Integration testing of the recruiter module confirms seamless collaboration between recruitment functionalities and AI-enhanced system components. The objective of these tests is to ensure that data flow, processing logic, and user interactions across interconnected modules remain consistent and reliable. Each test case validates that recruiter actions are correctly synchronized with backend services, databases, and AI-driven processes.

In IT-R001, the creation of new job posts by recruiters is verified by ensuring that job data is accurately stored in the database and immediately reflected on the recruiter dashboard. IT-R002 evaluates the interaction between the talent search module and the AI-based ranking engine, confirming that candidate lists are generated accurately and ranked according to recruiter-defined filters and relevance criteria. Test case IT-R003 focuses on portfolio data integration, ensuring that the recruiter dashboard loads complete academic and professional information for each candidate without missing or corrupting sections.

Furthermore, IT-R004 validates the integration between messaging and notification modules, guaranteeing instant delivery and proper synchronization of recruiter communications. Lastly, IT-R005 ensures that the export functionality retrieves the correct candidate datasets from the database and generates well-structured files suitable for offline analysis and reporting. Together, these tests confirm the stability, accuracy, and effectiveness of the recruiter module within the overall system.

#### Integration Testing 4: Portfolio and AI Engine

**Objective:** To validate that portfolio modifications adding/deleting content, switching templates, and auto-formatting trigger accurate AI analysis, ensuring that insights, skill gap calculations, and recommendations remain synchronized as shown in table 6.18.

**Table 6.18** IT 4 Portfolio and AI Engine

| Test Case ID | Test Description                                     | Input                         | Expected Output                             | Actual Output                               | Remarks |
|--------------|------------------------------------------------------|-------------------------------|---------------------------------------------|---------------------------------------------|---------|
| IT-PA001     | Add publication and generate insights                | Portfolio Module + AI Engine  | Insights updated with new metrics           | Insights updated with new metrics           | Pass    |
| IT-PA002     | Switch portfolio template and check AI compatibility | Template Engine + AI Engine   | Layout updates without affecting AI scoring | Layout updates without affecting AI scoring | Pass    |
| IT-PA003     | Delete a skill and re-run AI analysis                | Skill Module + AI Engine      | Skill gap recalculated accurately           | Skill gap recalculated accurately           | Pass    |
| IT-PA004     | Auto-format and sync with AI suggestions             | Formatting Module + AI Engine | Recommendations integrated into new layout  | Recommendations integrated into new layout  | Pass    |

This integration test examines the interaction between the portfolio builder and the AI engine, ensuring that user-initiated changes dynamically influence career insights and analytical outputs. The primary objective is to verify that data exchanged between the portfolio management module and the AI layer remains accurate, consistent, and synchronized in real time. These tests ensure that user actions on the interface are correctly reflected in backend processing and AI-driven decision logic.

In IT-PA001, adding a new publication triggers the AI engine to reanalyze portfolio content, recalculate skill relevance, and update academic performance metrics accordingly. IT-PA002

validates that switching between different portfolio templates does not disrupt AI scoring logic, keyword extraction, or metadata interpretation. Test case IT-PA003 confirms that when a skill is removed from the portfolio, the AI engine appropriately recalculates the skill gap and updates personalized recommendations. Finally, IT-PA004 demonstrates that auto-formatting the portfolio preserves all AI-generated suggestions and insights, ensuring seamless integration between visual design changes and underlying analytical processes.

## 6.2 Tools and Technologies

Table 6.19 provides a comprehensive overview of the key tools and technologies utilized throughout the development of the Customized Academic Portfolio project, clearly outlining their respective versions and specific purposes. The table includes a range of software tools used for documentation, presentation, and system design such as MS Word and PowerPoint for formal reporting and presentations, and Adobe Illustrator and Figma for creating user interface designs, wireframes, and visual assets. These tools support clear communication of ideas, visual consistency, and adherence to professional design standards across the platform.

In addition, the table highlights backend development and data handling technologies, including Python and the Django framework for implementing server-side logic, along with SQLite for development-stage data storage and Pandas and NumPy for efficient data processing and numerical analysis. AI and machine learning integration tools, such as the OpenAI API, NLTK, and TensorFlow/Keras, are also documented, reflecting their role in implementing intelligent features like text analysis, keyword extraction, and recommendation systems. These technologies enable data-driven decision-making and enhance the personalization capabilities of the platform.

Furthermore, the table lists tools for API testing and version control, including Postman for validating backend endpoints and Git with GitHub for source code management, collaboration, and version tracking. Finally, frontend technologies such as HTML, CSS, JavaScript, and Bootstrap are included, demonstrating their contribution to building responsive, accessible, and user-friendly interfaces. Collectively, these tools and technologies ensure a structured, scalable, and efficient development workflow, support maintainability and future enhancements, and enable smooth collaboration throughout the software development lifecycle.

**Table 6.19** Tools and Technologies

| <b>Tools And Technologies</b> | <b>Tools</b>      | <b>Version</b> | <b>Rationale</b>                                                  |
|-------------------------------|-------------------|----------------|-------------------------------------------------------------------|
|                               | MS Word           | 2019           | Documentation and report writing                                  |
|                               | MS PowerPoint     | 2019           | Presentation creation for project defense                         |
|                               | Adobe Illustrator | 2021           | UI design and graphics creation                                   |
|                               | Figma             | Latest         | UI/UX design and prototyping                                      |
|                               | Postman           | Latest         | API testing and integration                                       |
|                               | Git and GitHub    | Latest         | Version control and collaborative development                     |
|                               | MS Project        | 2019           | Gant Chart                                                        |
|                               | Python            | 3.12.1         | Primary backend programming language                              |
|                               | Django            | 5.0.4          | Backend framework for handling user data and portfolio generation |
|                               | SQLite            | Latest         | Lightweight database for                                          |

|  |                                  |        |                                                      |
|--|----------------------------------|--------|------------------------------------------------------|
|  |                                  |        | efficient data storage and retrieval                 |
|  | OpenAI API                       | Latest | AI-driven resume analysis and content generation     |
|  | NLTK                             | Latest | Text processing for AI-based resume parsing          |
|  | Pandas and Numpy                 | Latest | Data handling and AI model support                   |
|  | TensorFlow/Keras                 | Latest | AI/ML-based smart recommendations and resume parsing |
|  | HTML, CSS, Javascript, Bootstrap | Latest | Basic frontend for user interaction                  |

### 6.3 Summary

This chapter presents the various testing approaches applied to the Customized Academic Portfolio (CAP) platform to ensure system reliability, correctness, and overall quality. The testing strategy includes unit testing, functional testing, and integration testing, each targeting different levels of system validation. All major modules such as user authentication and login, portfolio builder, AI-based recommendation engine, recruiter tools, and student-specific features were systematically evaluated using well-structured test cases.

These test cases were designed to cover both standard operational scenarios and critical edge-case conditions to verify robustness under diverse user behaviors and data inputs. Through this comprehensive testing process, the platform's functional accuracy, module interoperability, and user experience were thoroughly assessed, ensuring that the system meets defined requirements and performs consistently across all use cases.

# **Chapter 7**

## **Conclusion and Future Work**



## **7 Conclusion and Future Work**

### **7.1 Conclusion**

The Customized Academic Portfolio (CAP) platform is designed to streamline the creation, management, and presentation of academic and professional portfolios, while offering intelligent insights to students, researchers, and recruiters. The primary objective of this project was to develop a comprehensive digital ecosystem where users can construct a professional academic portfolio, monitor achievements, and receive AI-powered recommendations, all within a single platform. For recruiters, CAP provides efficient tools to search, filter, and rank candidates, while students gain personalized career guidance and actionable suggestions for skill enhancement.

CAP has successfully transformed the portfolio-building process, making it more organized, interactive, and user-friendly. Traditionally, students relied on scattered documents, CVs, and manual updates, which were often inefficient and prone to errors. With CAP, all relevant academic and professional information is centralized, digitally accessible, and easily shareable through unique URLs or downloadable PDF formats.

A significant advantage of CAP is its intuitive and responsive interface. Users do not need technical expertise to navigate the platform. Whether managing portfolios, showcasing publications, or searching for talent, students, researchers, and recruiters can interact with the system effortlessly. The platform ensures seamless navigation, clear workflows, and a smooth user experience across all modules.

During development and testing, rigorous quality assurance was performed on features including profile creation, portfolio section management, publication synchronization, AI-driven recommendations, recruiter search functionalities, and PDF export. All modules have been thoroughly tested and operated as intended, ensuring reliability and consistency.

In conclusion, CAP effectively achieves its purpose of enhancing academic portfolio management while delivering intelligent, data-driven recommendations. It consolidates scattered information, simplifies portfolio creation, and increases visibility for academic accomplishments. By combining user-friendly design, AI-powered insights, and robust recruiter tools, CAP offers a holistic solution that is both practical and forward-looking. It represents a significant advancement in integrating technology into academic and professional development, making achievements more accessible, actionable, and impactful for all

stakeholders. The platform addresses existing challenges in portfolio management while laying the groundwork for future enhancements, including advanced analytics, personalized learning pathways, and improved recruiter-student matching, further cementing its role in the academic and professional landscape.

## **7.2 Future Work**

Although the current version of the Customized Academic Portfolio (CAP) platform is fully functional and successfully meets its objectives, there is substantial scope to enhance its features, usability, accessibility, and overall user experience. The future development of CAP aims to make the platform smarter, more interactive, and inclusive, while ensuring data security and seamless performance. The following initiatives are proposed for future work:

### **7.2.1 Mobile Applications (Android and iOS)**

Developing dedicated mobile applications for both Android and iOS platforms will allow students, researchers, and recruiters to access the Customized Academic Portfolio anytime and anywhere. The mobile apps will provide a fully responsive interface for portfolio creation and management, AI-driven career insights, real-time notifications for job postings and application updates, and quick communication between students and recruiters. Offline features, such as draft portfolio editing and cached job listings, can also be integrated to enhance accessibility in areas with limited internet connectivity.

### **7.2.2 Advanced AI Features**

Future enhancements will focus on expanding the AI engine to deliver deeper and more actionable insights. This includes personalized skill gap analysis to identify areas for improvement, automated candidate ranking based on multiple criteria including skills, experience, and academic achievements, and career path recommendations tailored to individual profiles.

### **7.2.3 Enhanced Portfolio Customization**

To improve engagement and presentation, users will be given more options to fully customize their portfolios. Features may include custom themes, dark mode, adjustable fonts and layouts, and interactive section designs that allow dynamic visualization of academic achievements and projects. Users will be able to create unique portfolio styles tailored to their professional branding, enabling greater flexibility for sharing with recruiters or academic institutions.

#### **7.2.4 Multi-Language Support**

Introducing multi-language support will make the platform accessible to a broader audience. In addition to English, languages such as Urdu and other regional languages will be supported, allowing users from diverse backgrounds to navigate the platform comfortably. This will also enhance inclusivity for students and researchers in multilingual contexts, increasing adoption and engagement.

#### **7.2.5 Strong Security and Privacy Measures**

Future work will emphasize advanced security and privacy protocols to protect sensitive academic and professional data. This includes two-factor authentication (2FA), end-to-end encryption for messages and file transfers, role-based access control, and compliance with data privacy regulations such as GDPR. These measures will ensure that portfolios, personal information, and communications remain secure from unauthorized access, maintaining user trust and platform integrity.

#### **7.2.6 Performance and Scalability Optimization**

As the platform grows in user base and data volume, performance optimization will be essential. Future improvements will focus on enhancing system speed, reducing loading times, and ensuring smooth responsiveness during intensive operations such as AI computations and portfolio exports. Scalability enhancements will allow the platform to handle a growing number of concurrent users, extensive portfolio datasets, and real-time analytics without lag or downtime, ensuring a seamless experience for all stakeholders.

#### **7.2.7 Integration with External Academic and Professional Platforms**

Future iterations may include integration with platforms such as LinkedIn, ORCID, Google Scholar, and research databases to automatically import publications, certifications, and professional experiences. This will further reduce manual data entry, keep profiles up to date, and enrich AI recommendations for both students and recruiters.

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